

Adaptive Adversaries in Byzantine-Robust Federated Learning: A survey

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Abstract

Federated Learning (FL) has emerged as a prominent paradigm for collaborative machine learning that enables model training without exposing private data through data decentralisation. Despite these advantages, FL systems remain vulnerable to significant security challenges affecting both privacy and robustness. This paper examines vulnerabilities in FL with a particular focus on model robustness, identifying critical gaps in existing defences against adaptive adversaries that alter their attack strategies after disconnection and subsequently rejoin the system. To the best of our knowledge, prior surveys have not explicitly addressed adaptive adversaries, motivating the focus of this work. We make five contributions: (1) a comprehensive overview of FL systems and their fundamental components, (2) a systematic review of vulnerabilities affecting FL, (3) an analysis of baseline attack vectors alongside state-of-the-art attack and defence mechanisms, (4) the introduction of a novel baseline attack based on Reconnecting Malicious Clients (RMCs), and (5) identification of future research directions for mitigating adaptive attacks. Experimental results demonstrate that widely used secure aggregation rules, including Krum and Trimmed Mean (commonly used as a baseline for other solutions to compare against), are insufficient against such attacks, and existing improvements to these methods fail to address this limitation. These findings highlight the need to reconsider FL security paradigms with explicit consideration of adaptive adversaries.

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Keywords: Artificial Intelligence; Machine Learning; Federated Learning; Secure Aggregation; Adaptive Adversaries; Byzantine-Robust Aggregation

1. Introduction

In recent years, Federated Learning (FL) [1] has surfaced as one of the leading paradigms for distributed learning, aiming to utilize multiple clients along with collaborative machine learning (ML) to avoid the need for exposing training data. Initially proposed ideas revolve around server-assisted architectures where a central server, also known as the aggregator, is the centralised anchor for FL systems, responsible for distributing, updating and securing the global model that clients contribute to [2, 3]. Motivated by privacy enhancements of data that the models are trained on [1], it is one of the leading solutions to the well-known limitations of large ML models using centralised data for training, requiring gathering all sensitive data in one place. Although it appears promising thus far, FL was found to carry its own set of security concerns affecting both privacy and robustness of the trained models [4, 5, 6, 7] with two widely-recognised types of adversarial entities: external and internal. External applies to all malicious actors intending to arbitrarily influence the model or reveal clients' data while not being a part of the FL system, and internal applies to all those that are.

Internal adversaries (i.e., insiders) possess a more complete insight into the global model and learning protocols, while being harder to detect, as such, making them a greater threat to FL systems [8, 9], which can be carried out by both the server and clients. Motivating researchers to further explore the extent of insider threats [10, 11, 12, 13], where recent literature most often categorises insider attacks into two groups, Byzantine and Sybil attacks [13, 14, 15, 16]. The former describes malicious, arbitrary updates which aim to degrade the performance of a model, and the latter refers to collaborative methods of attacks, either through joint efforts of multiple attackers or simulation of malicious clients [13]. These attacks then aim to influence models' behaviour or infringe the privacy of training data stored on the clients [17, 18, 19]. This prompted a response from the research community, which has caused two distinct types of defence mechanisms to emerge, either concentrating on the model-robustness against attacks on models behaviour [20, 21] or privacy-preservation, focusing on prevention of any inference of sensitive data stored on clients [22, 23, 24].

In this paper, we concentrate on aspects of robustness for FL security, which, once again, can be influenced by adversaries in one of two ways: data poisoning and model poisoning. Concentrating

on manipulating training data or model updates, respectively, to modify models’ performance. Although differing in implementation, poisoning attacks are described by some works as a special case of model poisoning attacks [15], as it follows that manipulation of data on which local models are trained results in the local model itself acting as a specific instance of a poisoned model. Going as far as to suggest that robust methods against model poisoning could work against data poisoning [15]. In general, poisoning attacks are noted to be either: (1) targeted for a specific label or a group thereof, or (2) untargeted, aiming to maximise degrading performance across all predictions [25, 26, 27, 28]. Notably, despite those being the main categories for attacks, we notice that these commonly used notions are somewhat incomplete, as recently a new type of attack has emerged where the adversary targets a source class to be misclassified but does not mind which target class it is misclassified as, so long as it maximises the chance of the misclassification of the source class, making it semi-targeted [29]. Further, there is yet another category known as backdoor attacks, often attributed with targeted attacks, with the main difference being the requirement for the model to behave as expected on all inputs until a backdoor is present, allowing the adversary to dictate the behaviour of the model [30, 31, 32].

Lastly, the potential of the attack highly depends on the adversarial capabilities, which are usually defined between black-box and white-box, about the attackers’ knowledge about the ML model and its surrounding system. In case of internal threats, it can be reasonably assumed that the adversary operates as a white-box attacker, given how they will get insider knowledge of the FL system and receive the global model along with the updates [13, 32, 33]. This ever-present threat of adversaries possessing white-box knowledge about the system keeps opening new doors for potential attacks while having many methods to come by undetected. One is the possibility of an adversary reconnecting malicious clients to the system after being forcibly disconnected due to low reputation during the learning process, allowing them to adapt their malicious clients and reconnect to the system, aiming to continue the attack and significantly increasing its severity. To our knowledge, despite published literature present in domains of reconnection-efficiency [14, 34, 35] and reputation-based FL for malicious behaviour detection [36, 37, 38, 39], there is no consideration for such attacks, leaving them unexplored. Further, recent literature acknowledges no commonly accepted formal definition for “adaptive adversaries” [40], highlighting the need for further work in this domain.

Having said that, after initially scoping the field of FL security, specifically in the domain of

model-robustness, we found no prior studies extensively covering and focusing on the threats posed by adaptive adversaries. This also includes the lack of baseline evidence for the security risks they cause. Other surveys in the field of FL security usually concentrate on detecting specific attack methods and excluding adversaries from global model computation. Even works such as [41] covering trustworthy artificial intelligence in FL overlook this aspect. This gives us motivation to further explore this critical element of FL security and serve this work as a baseline evidence for this threat.

In this paper, we aim to bridge this gap and bring attention to this form of threat, providing reasonable concern and motivation for their standardisation and prevention. As such, the contributions of this paper are as follows:

1. We outline and summarise the domain of FL systems and FL security, categorising its founding elements.
2. Extensively overview the existing vulnerabilities with emphasis on model robustness, providing readers with the necessary background for present state-of-the-art attacks and pointing out known FL security needs.
3. Present modern defence methods and their approaches to counter these attacks, and comment on their differences in approach.
4. Introduce a novel method of adaptive adversarial attack using client reconnections, where we coin the term Reconnecting Malicious Clients (RMCs), conducting experiments on its effectiveness against some well-known defence mechanisms.
5. Research directions that will provide further pointers on how to counter this method with promising effectiveness. Encouraging the re-evaluation of current state-of-the-art methods that would prevent the effectiveness of reconnected malicious clients.

It is worth noting that the direction of this work is structured first to provide a comprehensive survey of the security landscape within Federated Learning (FL), contextualising both foundational and emerging challenges in this distributed machine learning paradigm. The initial sections will cover the breadth of FL Security, beginning with a precise definition of FL and its system architecture, followed by a systematic categorisation of known attack vectors, including poisoning and inference attacks. It will then examine the range of adversarial models encountered in literature, distinguishing between internal and external threats alongside white-box and black-box assumptions. Existing

defensive mechanisms spanning robust aggregation methods will be critically analysed regarding effectiveness, assumptions, and limitations.

After establishing the broad landscape, the paper narrows its focus to a specific gap identified within adversarial approaches: the potential occurrence of Reconnecting Malicious Clients. The strategic behaviour of adversarial participants who rejoin the training process with altered attack strategies to evade detection and amplify attack efficacy. While much of the existing literature assumes a static presence or absence of malicious clients, this dynamic behaviour remains under-explored, despite its growing relevance in practical deployments of FL.

By adopting this breadth-to-depth structure, the paper not only surveys the current state of Federated Learning security but also contributes novel insights by highlighting and investigating a previously under-represented threat model. This approach enables the reader to situate the specific contribution within the wider body of FL security research, while being friendly to all readers unfamiliar with the domain of FL security.

With that said, the rest of this survey is organised as follows. Section 2 will briefly discuss details on the collection of the literature for this study. Section 3 will then present a literature review and preliminaries, segmented to cover separate founding elements of FL, including its definitions and architecture, security vulnerabilities, and defences; this section will also serve as an entry point for our novel attack method, where we will provide exact definitions for Reconnecting Malicious Clients and Adaptive Attacks. Section 4 will guide the reader through our research methodology, where we will discuss the approach taken and experimental setup. Section 5 will then discuss the results of experiments conducted based on the surveyed attacks and mechanisms, establishing the baseline effectiveness of the known defensive measures. We will also present the effectiveness of our discovered method of attack and compare it to the known approaches. Then, based on our findings, we propose potential directions of research in this domain in Section 6. Finally, this paper will be concluded in Section 7, reviewing our findings and summarising our results.

2. Literature Collection

The literature review process was conducted with the primary aim of capturing the current state-of-the-art in Secure Aggregation protocols, byzantine-robust federated learning, and the specific implications of threats originating from adaptive adversaries. To ensure both breadth and relevance, the methodology relied on a structured querying process using Google Scholar as the primary search

engine. IEEE Xplore, Scopus, and arXiv were the most accessed databases for peer-reviewed works and preprints in the case of arXiv.

Queries were systematically constructed to reflect the thematic focus of the study. They included combinations of core terms such as "secure aggregation", "federated learning", "adaptive adversaries", and "byzantine-robustness" among others. Boolean operators (AND & OR specifically) alongside exact phrase matching were employed to refine results and exclude non-relevant domains. Each query was executed iteratively, with the top 50–100 results being assessed based on their relevance to the topic, as determined by title, abstract, and journal in which the literature was published in.

The initial filtered search yielded a total of 97 Google Scholar results combining all of the above mentioned terms, 23 of which came from arXiv pre-print database, 10 from IEEE Xplore, and 17 from Scopus. We focused on the most relevant works in their respective categories, as a partial aim of this study is to present to the reader the breadth of the FL security domain in a manner friendly to those unfamiliar with the field.

To maintain comprehensiveness, backward and forward citation tracking was also performed on key foundational papers identified in the initial query stages. This helped identify influential but under-indexed works that might have been missed by keyword searches alone. Additionally, peer-reviewed articles, conference papers, and formally published technical reports were priorities for inclusion. However, given the rapid development of the FL security domain, we decided to include preprints if their quality and significance were comparable to that of the peer-reviewed literature.

3. Preliminaries

In short, Federated Learning (FL) is a framework that generally aims to decentralise the data on which the model is trained. Although the exact implementations may vary strictly depending on the extent of data heterogeneity stored on the clients (also known as data-owners) and resources available to the system and clients, all architectures will share this quality, making it a defining feature of FL. For clarity, we provide a better understanding of concepts in FL research and how they are categorised in this section.

3.1. Federated Learning: Definitions

Ensuring readers' coverage of FL fundamentals, we start by briefly describing all elements involved and then present how they are all connected in the most commonly seen architectures. As such, FL consists of the following:

Aggregator - Entity responsible for aggregating all local models, producing a global model. Often-times it can be described as a server within the literature, handling processes like global security of the system, global model initialisation and distribution. Should only see local model updates with no more information about the underlying data they were produced with.

Client - Owner of the data, clients are entities that produce their own local models using their private data, which FL systems are trying to protect, and send them to the aggregator for further computations. The sent and received updates should not give other entities an option to reveal privately stored data.

Local Model - Models trained by clients strictly on their limited data that they store privately, serving as updates for the global model. Local models and their effectiveness will highly depend on the private data stored and the clients' computational capabilities. As such, individual clients' contribution to FL systems is often considered based on those factors.

Global Model - Final model that the FL systems end up with at the end of each round. The computation of global model is done through aggregation, where all models are "combined" together to generate the final model that will be once again redistributed among clients as updates or as the final model.

Round - Also referred to as a FL iteration, it encompasses the iterative process of redistributing global model to clients, receiving all the local model updates and aggregating the global model.

As previously mentioned, FL aims to decentralise data to provide greater security, such that the data owners would not be required to transfer their data over the network. Instead, they each individually train their own local model that later contributes to the global model [2, 42]. Works describing practical applications, such as [43], formalise the process in the following way.

Assume N clients $\{C_1, C_2, \dots, C_N\}$ store their respective datasets $\{D_1, D_2, \dots, D_N\}$. Each client possesses its own localized learning model $\{L_1, L_2, \dots, L_N\}$ with r parameters (weights), where each L_i can be expressed as $\{W_1, W_2, \dots, W_r\}$. Following the aim of the FL framework, the goal is to iteratively train a global model G using local models without accessing the individual

datasets belonging to the clients. The process begins by distributing an initial G among all clients, establishing a universal set of r parameters. Then, for each iteration t (also called a round), every client C_i trains their local model using the initialized G over their dataset D_i , resulting in N sets of parameter updates $\{\hat{W}_1^t, \hat{W}_2^t, \dots, \hat{W}_r^t\}$. After computation of all local models, the global model G^t and its parameters are then aggregated using an aggregation rule Δ and redistributed among all clients, with all their local models L_i updating as visible on Equation (1):

$$L_i^{t+1} \leftarrow G^t \leftarrow \Delta(\hat{W}_1^t, \dots, \hat{W}_r^t) \quad \forall i \in \{1, \dots, N\} \quad (1)$$

This process is then repeated until the global model reaches a satisfactory level of performance.

3.2. Federated Learning: Categories

FL implementations may differ depending on the data heterogeneity and structure of the FL system. As such, we highlight the most commonly used categories in relation to those properties.

3.2.1. FL Architecture

Properties of a FL system that describe relationships between entities and their roles involved in FL process are defined by the architecture, which are categorized into three main groups: Centralized, Decentralized and Clustered.

Centralized. (Server-client) These types of architectures are characterised by the presence of a central entity, which usually is a server, with the responsibility of distributing and updating the global model among all connected clients [44]. The central server, also referred to as an aggregator, acts as a main control point for the entire FL system [45]. Often resembling star topologies as all clients are connected to only one central entity. Figure 1 presents a centralised architecture in FL. The overall process depicted follows the basic concepts that have been discussed already. First, the aggregator initialises the global model to clients, who follow up with training their own local models and uploading them back to the aggregator as updates. Then, the aggregator computes the global model based on the updates and repeats the process. In relation to other architecture types, centralised FL is the least complex for implementation, making it the most common architecture for FL.

Decentralized. (Peer-to-peer) A variant of FL that aims to eliminate the need for a central server as visible on Figure 2, ergo, removing the single point of failure. Here, every client has

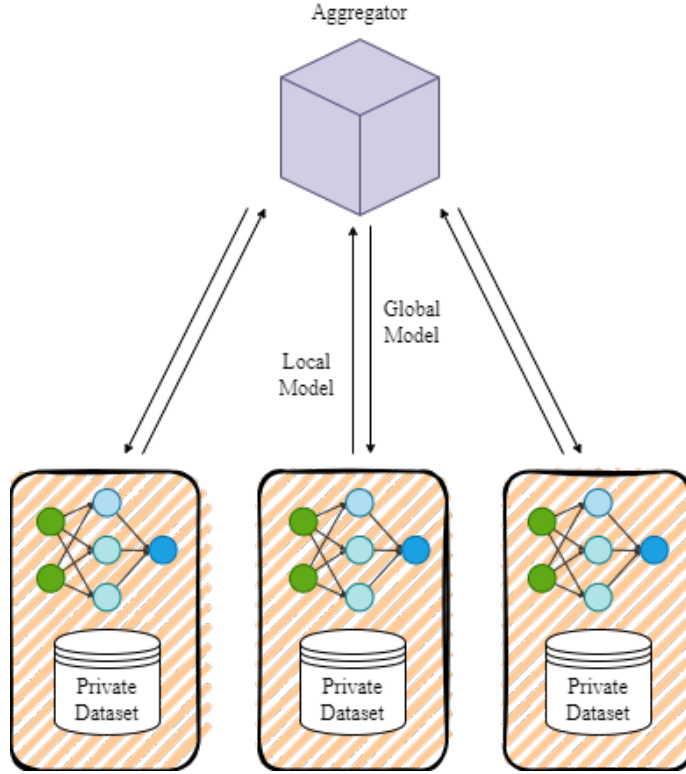


Figure 1: Centralized Federated Learning overview. The central server initialises the model to all clients, which then proceed to train their local models and upload them back to the server as an update. After aggregation, the server redistributes them back to the clients. This process repeats until a satisfiable model performance is reached.

access to their own data as well as the aggregated model updates from other clients, meaning that clients distribute the model updates between themselves [46, 47, 48, 49]. Figure 2 also presents how local training is performed among peers in the network; instead of uploading local updates to the aggregator clients, they try to approximate its behaviour using some arbitrary decentralised mechanism. Clients’ choice of peers to share their updates with often depends on the topology of the network; as such, it is worth noting that clients will often distribute their local updates to a subset of their peers rather than broadcasting them to all other clients. Interestingly, unlike in centralised architectures, here each client is allowed to incorporate their own aggregation strategy for model updates [50]. Making it more difficult to poison, especially when a poisoning strategy is adapted for a specific aggregation rule like Full-Krum or Full-

Trim [51]. However, the trade-off for security benefits in decentralised architectures is increased implementation complexity and higher difficulty in handling data heterogeneity [49].

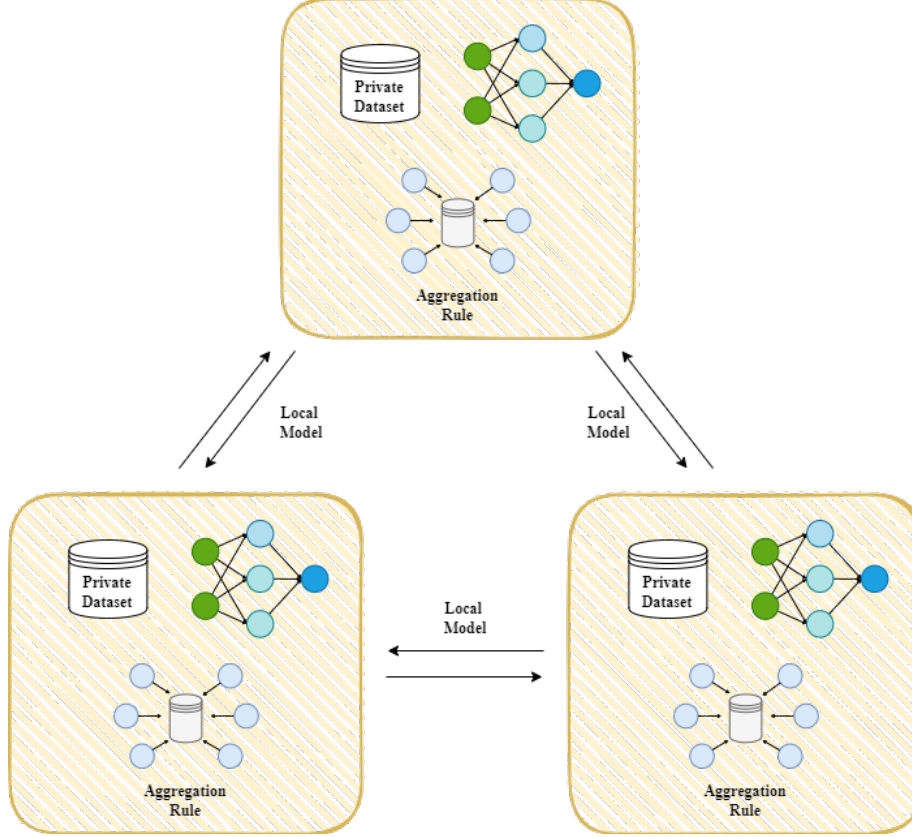


Figure 2: Decentralised Federated Learning overview. Clients collaborate without any central entities to produce a global model by mutual exchange of their trained local parameters.

Clustered. Framework addressing data heterogeneity issues in FL. As presented in Figure 3, it aims to improve models' performance by grouping clients that store data with similar properties, such as distribution, feature space or sample space. Proceeding to train separate models for each cluster, adapting the cluster aggregation strategies to a specific group of clients and then aggregating cluster models, producing the global model [52, 53]. The process from Figure 3 may be described as follows. Aggregator initialises the model to all known clusters within the network, which subsequently is distributed among the clusters themselves. Clients, as per usual, compute their own local models based on their privately stored data. However,

once the local models are computed, prior to the transfer of local updates to the aggregator clusters, oftentimes compute their own collective update of the cluster, known as a cluster update. That then is uploaded back to the aggregator, which computes the global model and once again distributes it among clusters, repeating the process. Clustered FL comes as its own category due to the fact that it concentrates on the element of client grouping, coming in both centralised and decentralised variations depending on the architectural demands. It was shown to display mathematical guarantees for performance, which involve general non-convex objectives [54], making it equally applicable to real-world scenarios as other architecture types.

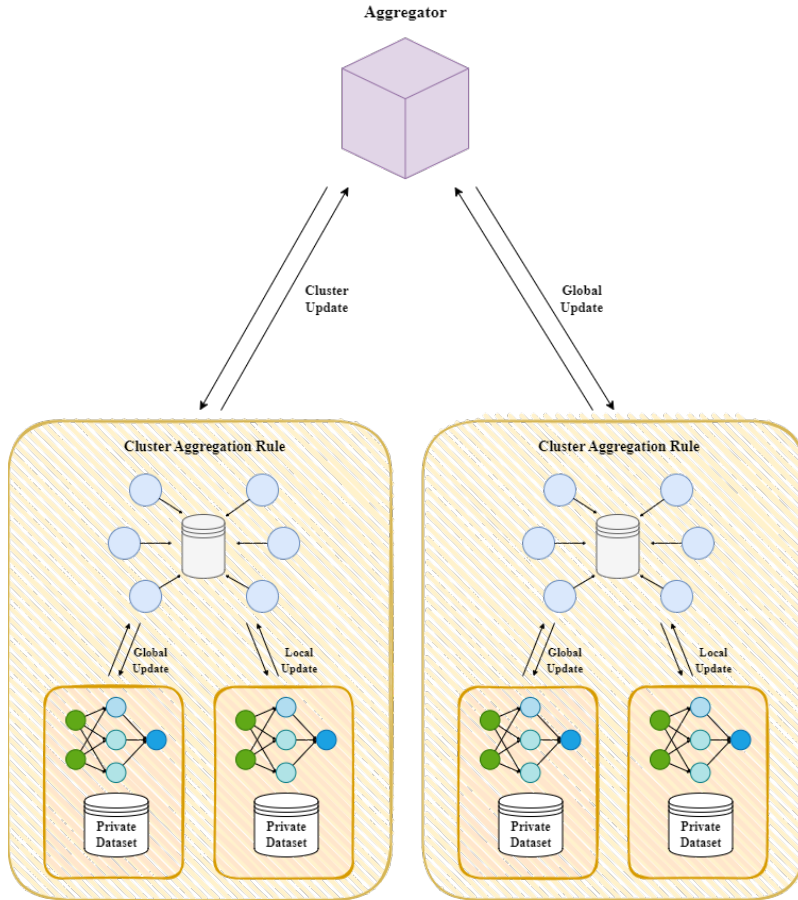


Figure 3: Clustered Federated Learning overview. Clients are grouped into clusters based on arbitrary properties and then compute cluster models that are then collectively aggregated.

3.2.2. Data Heterogeneity

Differences in data structure among clients in FL are a common occurrence in practical scenarios. As such, it is commonly expected that data heterogeneity will influence the implementation of FL systems depending on the type and extent of those differences [55, 56, 57].

Horizontal FL. (HFL) When clients possess datasets with similar or even identical feature space yet differing sample space, we can implement it such that the models trained base their computations on the shared feature space. In essence, it is like training the same model but with different datasets, which we refer to as horizontal FL [58]. To our knowledge, this is the most commonly used form of FL especially present when implementing dee learning through FL as HFL allows for a significantly larger sample size among clients [58].

Vertical FL. (VFL) Reflection of HFL, where clients possess very similar sample space but differ in the feature space. VFL operates on different training protocols, which account for those differences. To be specific, clients instead of sharing their local models for aggregation, they share direct outputs of their trained model based on the stored data, also known as intermediate results [59], which are then used for computation of the global model. The VFL process is often coordinated by one of the clients, also called an active party, rather than the aggregator. Nevertheless, due to the specificity of the VFL scenarios, the complexity of their implementation increases when ensuring models' performance and security [60, 61].

Federated Transfer Learning. (FTL) In a scenario where clients possess completely different datasets, such that both sample space and feature space differ, FL may adopt transfer learning to adapt itself to this circumstance. FTL leverages cross-domain insights between clients to attempt generalisation, mitigating issues regarding different data distributions [62]. However, FTL is characterised by unique challenges in the FL domain, as it is less straightforward to allocate client contributions or interpret models given the differing data distributions across all dimensions [63]. Only amplifying the extent of those limitations more than other architectures do.

3.3. Security Vulnerabilities

As previously established, security of FL systems concerns two main elements, most vulnerabilities are a subpart of: privacy and robustness. Collaborative and data-decentralised training introduces security vulnerabilities to both. Although this survey concentrates on robustness in FL

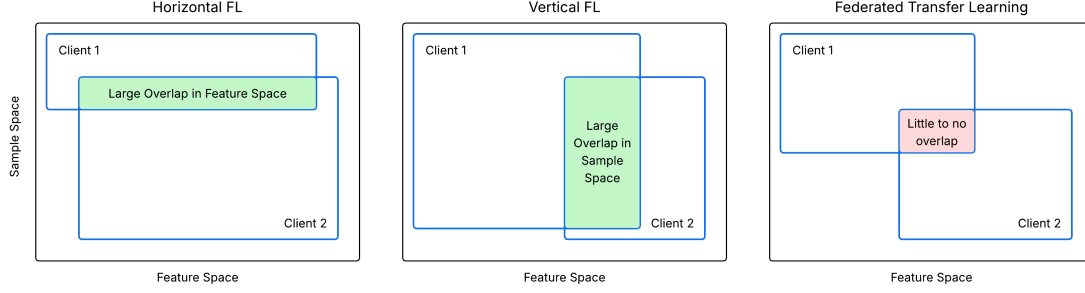


Figure 4: Visualisation of different approaches to FL depending on the data heterogeneity.

security, for completeness, we briefly discuss other elements of the FL security landscape to give the reader a broader perspective for this survey.

A foundational distinction in the study of adversarial threats to machine learning models, including in FL contexts, lies in the attacker’s knowledge and access to the target model. This categorization broadly defines two primary threat models: white-box and black-box attacks.

White-box. These occur when the adversary has full access to the model’s internal parameters, architecture, and training procedure. This allows for precise crafting of adversarial examples or extraction of sensitive information, often leveraging gradient-based optimization to exploit model vulnerabilities.

Black-box. In contrast to white-box attacks, black-box attacks assume no internal access to the model; the adversary can only observe the output in response to input queries. These attacks rely on techniques such as transferability from surrogate models or iterative query-based methods to infer model behaviour and generate adversarial inputs.

It is worth noting that in case of FL security threats, it is commonly assumed that the adversaries have white-box knowledge about the global model they are targeting due to the significance and likelihood of insider threats that exist in FL environments.

3.3.1. Privacy

The main target of privacy threats for FL is the data stored by the individual clients, as such, adversaries aim to cause privacy-leakage. As of now there are three main categories of potential methods that adversaries can use to achieve this goal.

Model Inversion. FL is vulnerable to any form of inversion of data from models outputs

or parameters, as is the case with model inversion attacks, where an adversary tries to directly reconstruct the private data of other clients using their access to this information [64]. Consequently, reverse engineering general patterns from the private datasets exposing their properties [65, 66, 67]. These attacks can be performed both in black-box and white-box adversarial settings, as the adversary needs at least access to the model predictions which they are going to inevitably receive during the training process.

Gradient Inference. Focusing specifically on gradients, this method focuses on the gradients during model updates as they are exchanged between server-clients in case of centralized FL or between clients themselves in case of decentralized FL. It is capable of reconstructing specific datapoints, marking it as a significant threat to FL with this extent of privacy-leakage [68, 69, 70, 71, 72]. Nevertheless, these attacks often require white-box adversarial capabilities, as they need to access other clients’ gradients directly.

Membership Inference. Adversaries may want to establish whether a specific data record exists within one specific or any of the clients datasets using models outputs [72, 73, 74, 75]. These attacks are especially concerning in medicine related scenarios such as cancer treatment, as membership inference attacks enable the adversary to deduce whether specific sensitive healthcare information is present about an individual [76]. However, similarly to gradient inference attacks, membership inference requires white-box knowledge about the target FL system in order to be performed [77].

GAN-Based. Generative Adversarial Networks (GANs) utilize two separate neural networks that compete against one another. One generating synthetic samples of data attempting to mimic real datasets and the other evaluating their authenticity [78]. Intuitively, this concept may be used to reconstruct the data from other clients by using the global model to gain knowledge for adversarial examples representing privately stored data on other clients [79]. Once again however, works exploring this method assume white-box capabilities from the adversary, given how it requires full knowledge of the target model kept as a replica [80, 81, 82].

3.3.2. Robustness

In federated learning, ensuring models integrity means providing a guaranteed level of performance when targeted by an adversary. As such, vulnerabilities associated with model-robustness refer to all attacks that degrade models’ performance. Poisoning serves as the one and major vul-

nerability regarding this aspect, which within of itself can be segmented into two categories: source of the poisoning and its target. Adversaries targeting FL environments have two main points of entry for their attacks. It is either through data stored on malicious clients or directly through the model updates.

Data Poisoning. Since an adversary is usually assumed to be an internal threat as a part of the FL system, it directly allows them to modify their own dataset such that it will manipulate models’ behaviour and lead to decreased performance [83, 84, 85, 86]. Interestingly, data poisoning attacks can be viewed as a form of model poisoning attacks given that the attack poisons data, producing a poisoned local model, which subsequently poisons the global model, being the main target of the poisoning attacks [15]. This is unique to federated machine learning, as the poisoned data doesn’t have an effect on the global model directly.

Model Poisoning. Adversaries may choose to inject a poisoning attack directly on the local model itself to poison the global model. As an adversary trains its own local model they can introduce modifications to its parameters before passing it to the global model. Given that their local model will be initially trained on the received global model, further modifications can be adapted such that it doesn’t deviate as much from the global model in an attempt to not get detected but is effective enough to still prevent convergence and degrade models’ performance. This appears to be a more common choice for poisoning due to the fact that it gives an adversary greater control over the method of attack directly affecting the global model as it is presented by recent works [39, 49, 51, 84, 87, 88, 89, 90].

Impact of the adverse effects caused by poisoning attacks are often categorized through the target an adversary will choose for their attack. Which may vary depending on their aim, often considering aspects like detectability and severity of the attack. This survey identifies four main groups of attacks based on their targets and relevance in the surveyed literature, which go as follows.

Untargeted. Simplest and most severe overall, as those poison attacks aim for maximised loss of the model, minimising its accuracy across all classes. Due to the severity of these attacks, they usually end up as the easiest for detection as the results of the attack are blatant [90].

Targeted. Adversaries aiming to misclassify a specific source class as a specific target class without degrading performance across other classes. This increases the complexity of the attack but at the same time makes it harder to detect, given the specificity they aim for [6, 8].

Semi-Targeted. Adapted for scenarios when the adversary wants to misclassify a specific source class but does not specify the target class that they want it to be misclassified as. Misclassification to multiple target classes is deemed as satisfiable by an adversary [29].

Backdoor. These attacks enable the adversary to implement the named “backdoor”, that being a set of inputs or a specific pattern thereof, only after which the model will begin to degrade its performance. It is required for backdoor attacks not to affect the global model’s performance unless the backdoor is present, meaning that the model behaves as expected. Consequently, making these types of attacks hardest to detect, and giving the adversary the most control over the attack’s behaviour in real time [91].

With that said, we have now covered all relevant FL vulnerabilities in the surveyed literature. From this point onwards, we will further narrow down to the pre-established scope of this paper and extensively inspect attack vectors against model robustness along with state-of-the-art defence mechanisms in FL security.

3.4. Attack Vectors

Previously discussed security vulnerabilities may be exploited in more ways than one, and thus far, this work has only covered the general landscape of FL security. However, specific angles that an adversary may target the FL model from, known as attack vectors, are what define the extent of the severity of those vulnerabilities. Consequently, there is a constant need for further enhancements for defences that will counteract those attack vectors. In this and the upcoming section, we go through seven attacks total targeting model-robustness, three of which will be used for experimentation, other four will be present for completeness as they commonly appear within literature, however they often target a specific defence mechanism, scenario, or FL architecture that can’t be generalised for our experiments. The same applies to defence mechanisms, where we consider two for experimentation and include two more for completeness.

In FL security, attack vectors against model-robustness most often concentrate on preventing convergence either through data-to-model or model-to-model approach [92]. Most surveyed literature also shows that the most common attack type showing greatest adversarial potential in this context is Byzantine attacks [51, 93, 94], operating under the assumption that the FL system involves malicious clients that do not follow the specified protocol.

3.4.1. Flipping Attacks

Attacks that concentrate on changing the sign of either model updates or data, with or without specific conditions, will belong to this category. We provide the following list for most commonly listed flipping-based attacks.

Sign-Flip. A simple attack that flips the sign of each parameter update which automatically deviates them from benign clients. One may consider use of probabilistic methods to make the attack less detectable [95] although the underlying approach stays the same.

Label-Flip. A data poisoning attack that an adversary applies on their training data, by flipping labels of training samples before training their model, producing a poisoned model which deviates from one produced by benign clients [51]. Given L as the total count of classes, it is simply done by taking label l and flipping it to $L-l-1$. Interestingly, in binary classification tasks, label-flip attack can be viewed as data-oriented variation of sign-flip attack, as it has the same result on the poisoned training samples as sign-flip has on the poisoned model parameters.

Inner Product Manipulation. (IPM) Most sophisticated out of all sign-based attacks. Authors of [96] revealed a concrete vulnerability when using Stochastic Gradient Descent (SGD) algorithms. When aggregating global model parameters, the local model vectors and true gradient vectors are required to face the same general direction for convergence, which in turn makes an inner product between them positive. As such, the IPM attack aims to prevent that by making the inner product negative and preventing convergence. The authors describe the condition of this attack with the following expression:

$$\langle g, E[\Delta(V \cup U)] \rangle < 0 \quad (2)$$

Where g is the true gradient, and $E[\Delta(V \cup U)]$ is the expectation using aggregation rule Δ on a global model update using unified set of genuine gradients V and byzantine gradients U .

Bit-Flip. Targeting floating numbers within data, flipping their bits causing their values to sway the poisoned model away from genuine predictions. This method of attack is often considered as realistic as it can be carried out not only intentionally but also through hardware failure [92, 97].

3.4.2. Parameter-Based Attacks

Similarly, attackers may utilise a simple idea, which is to inject a perturbation of some sort into the local model updates or data that prevents the global model from converging. This can be done by introducing noise to the parameter updates or by arbitrarily choosing values for the updates. Both are present in the following list.

Gaussian-Noise. These attacks simply introduce randomly chosen noise to the parameters using a Gaussian distribution [93]. Interestingly, the noise attack in general can be modified to use different distributions or optimised using fixed vectors to maximise the effectiveness of this attack [98].

A Little is Enough. (ALIE) A more sophisticated variation of noise attacks as the choice of the noise introduced becomes bound to a range for each coordinate based on benign clients. The range specified by the authors who first introduced the attack [99] lies on an interval $(\mu - z^{max}\sigma, \mu + z^{max}\sigma)$ which limits the values of the noise, however it results with an attack that is much harder to detect by defensive mechanisms.

Same-Value. Simple attack which only aims to replace the model update to a single value across all parameters [92].

3.5. Defence Mechanisms

The vast majority of defences developed against the aforementioned attack vectors are either cryptography-based or statistics-based. In case of model-robustness, the most common point of defence lies in secure aggregation, and as such, we will be concentrating on this mechanism in our experiments. However, this paper also presents other mechanisms besides secure aggregation for completeness of this survey, so that the reader is aware of other points of interest in this domain.

3.5.1. Secure Aggregation

Secure Aggregation (SA) is a statistics-based protocol designed to protect the aggregation process by inspecting individual contributions. This is an arbitrary process as long as it is capable of excluding updates from malicious clients. Often, SA protocols assume that the majority of the clients that contribute to the global model are benign [20, 93], allowing for the statistical models to inspect the tendencies of updates, and those that deviate from the majority may be deemed as malicious. Researchers also look at the direction of received vectors for aggregation to evaluate

which ones are malicious or deviate from the majority under this assumption [100], or alternatively, the aggregator computes its own model and compares updates before aggregation [101].

In addition to these statistically-driven techniques, many Secure Aggregation protocols are augmented with reputation mechanisms to further reinforce robustness. Reputation systems maintain historical profiles of client behaviour, assigning dynamic trust scores based on consistency, alignment with global objectives, or conformity to expected statistical norms. For example, a client whose updates frequently deviate from the global model direction or degrade convergence performance may be assigned a lower reputation, reducing the weight of their updates in future aggregations or excluding them entirely, which then leads to eventual disconnection of the entity from the system. Conversely, clients with a history of contributing stable and beneficial updates receive higher reputations, thus having more influence in shaping the global model. This dynamic adjustment helps mitigate long-term threats such as Byzantine or passive poisoning attacks, where adversaries behave benignly in early rounds before launching a more impactful deviation.

Beyond reputation and directional statistics, recent research has also explored the use of hybrid techniques within SA frameworks. These methods combine multiple statistical metrics (such as norm-based thresholds, angle-based similarity measures, and clustering validity indices) to perform a more holistic assessment of client updates [102, 103]. Some protocols even incorporate machine learning models to predict the trustworthiness of updates based on past behaviours and feature representations of gradient vectors. This trend toward adaptive, learning-based SA reflects a broader shift toward more context-aware aggregation strategies that evolve in response to observed attack patterns and data heterogeneity.

The following is a list of SA algorithms that we often came across in surveyed literature, all of which present a different approach to SA.

Krum. Byzantine-resilient aggregation rule for SGD [104] which selects vectors from updates that are closest to $n - f - 2$ neighbours, where n is the total number of workers and f is the number of Byzantine clients. We may express the aggregation rule as follows:

$$Krum(V_1, \dots, V_n) = V_i \quad V_i = \operatorname{argmin}_i[s(i)] \quad (3)$$

Where V_i is the chosen vector from $V_1, \dots, V_n$ based on the minimised score value $s(i)$, which within of itself is defined over a sum of squared Euclidean distances for every worker i with closest workers j [104], as visible on Equation 4.

$$s(i) = \sum_{i \rightarrow j} \|V_i - V_j\|^2 \quad (4)$$

Trimmed Mean. Operates on the fact that one can get rid of statistical deviations that don't follow the majority by trimming both the highest and lowest update values before aggregating. Authors introducing the aggregation [105] describe that each update on coordinate k of received vector g can be expressed as:

$$g_k = \frac{1}{(1 - 2\beta)m} \sum_{x \in U_x} x \quad (5)$$

With β being a fraction of highest and lowest values between $[0, \frac{1}{2})$ to be trimmed, m the total number of clients and U_x the set of values after trimming.

Clipped Clustering. A state-of-the-art aggregation function proposed by [93] that combines centered-clipping and clustering aggregation methods. The authors use centred-clustering [106] that iteratively updates vectors based around the centre of all values to be aggregated, clipping them within a range such that it avoids inclusion of malicious updates that deviate too much from the centre. Thus, as described by [104] centred-clipping algorithm computes:

$$V_{l+t} = V_l + \frac{1}{n} \sum_{i=1}^n (x_i - V_i) \min(1, \frac{\tau}{\|x_i - V_i\|}) \quad (6)$$

This, as noticed by [93] can be combined with clustering aggregation schemes. Calculating pairwise cosine distances between the updates, which further segments the clients by computing those distances as follows:

$$\alpha_{i,j} = 1 - \frac{\langle \Delta_i, \Delta_j \rangle}{\|\Delta_i\| \cdot \|\Delta_j\|} \quad (7)$$

After which the largest group is taken for aggregation. Authors of clipped clustering combine these two by performing clipping prior to clustering, resulting in a simplified formulation:

$$\Delta_k \leftarrow \Delta_k \min(1, \frac{\tau}{\|\Delta_k\|}) \quad (8)$$

FLTrust. Ensures security of aggregation by introducing Trust Scores (TS) to every client update g_i which is based on how closely the updates are aligned with the servers update g_0 . The

authors of FLTrust [101] use *ReLU* activation function over pairwise cosine distances between the client and server updates to determine the TS for every client i .

$$TS_i = \text{ReLU}\left(\frac{\langle g_i, g_0 \rangle}{\|g_i\| \cdot \|g_0\|}\right) \quad (9)$$

FLTrust then normalises the client updates to the servers' magnitude by computing $\bar{g} = \frac{\|g_0\|}{\|g_i\|}$ and finally aggregates the trust-weighted average in the following way:

$$g = \frac{1}{\sum_i TS_i} \sum_i TS_i \cdot \bar{g}_i \quad (10)$$

Finally applying learning rate to the global update:

$$w \leftarrow w + \alpha \cdot g \quad (11)$$

3.5.2. Differential Privacy

Mathematical framework used for enhancing privacy-preservation in FL systems ensuring that the output received by the model does not reveal too much information about any individual record in the dataset [107]. This is done through controlled noise added to the dataset. Formally, differential privacy states that given a randomised algorithm M , it satisfies (ϵ, δ) -differential-privacy for all adjacent datasets D and D' if for all possible outputs S following inequality holds true:

$$P[M(D) \in S] \leq e^\epsilon \cdot P[M(D') \in S] + \delta \quad \forall \epsilon, \delta \geq 0 \quad (12)$$

Where ϵ controls the privacy guarantee, the closer it is to 0 the stronger the noise and subsequently the privacy of the mechanism. This inevitably results with more disrupted computations, as such choosing ϵ faces further complications regarding its applicability in practice [108]. On the other hand, δ defines the probability where two adjacent datasets exceed e^ϵ which in turn should be cryptographically negligible [107].

3.5.3. Secure Multi-Party Computation

Secure Multi-Party Computation (SMPC), also known as Multi-Party Computation (MPC), is a cryptographic framework that allows for joint computation between n parties without revealing their inputs. In MPC, inputs are generally divided into shares using schemes such as additive secret sharing [109] where a secret s is split into n random shares $s_1, \dots, s_n$ such that:

$$s = \sum_{i=1}^n s_i \pmod{p} \quad (13)$$

Where p is a prime. In FL environment, each client receives one share and, in its simplest variation, no subset of fewer than n clients can reconstruct s . SMPC enables FL to secure information about the system and other clients even if some parties are malicious or semi-honest (also known as honest-but-curious), which follow the protocol but attempt to infer information from it without causing disturbances.

In essence, SMPC protocols aim to compute $f(x_1, \dots, x_n)$ in a way that it achieves indistinguishability when a set of adversarial clients $A \subseteq \{C_1, \dots, C_n\}$ jointly attempt to observe outputs of the model and tries to learn information about the private inputs. SMPC is somewhat known as an "umbrella" term that encapsulates any solution that involves multiple parties necessary for secure computation. An example directly from FL literature may be found in [110] where authors obfuscate gradient updates such that the adversary would be unable to derive any information about the clients private data.

3.6. Adaptive Adversaries

3.6.1. Reconnecting Malicious Clients (RMCs)

Throughout the surveyed literature for FL security in the context of model-robustness, we discovered that there was no mention of Byzantine or Sybil clients reconnecting to the FL systems even after being forcibly disconnected or rejected. This includes extensive works reviewing trustworthy AI in relation to FL, such as [41], which lacks consideration for this type of threat. We bring this up in the context of FL as client trustworthiness and reputation systems are one of the crucial elements to the robustness of global models. At the centre of this problem lies client selection and control, which should enable FL environments to verify and exclude malicious clients that have been detected.

In practical scenarios, FL systems would try to disconnect detected malicious clients from the system, such that an adversary would not be able to modify the attack in real time once they notice their attack is ineffective, as it can be viewed in Figure 5. Currently, security mechanisms indirectly assume that once a client has been detected and disconnected from the FL system, it will either not reconnect or is guaranteed to be rejected once again. Recent research points out similar remarks in relation to adaptable adversaries [111, 112]. However, we also note that the works in this direction

so far have been scarce. This assumption only holds true for FL systems that are not dynamic or reconnection-efficient, essentially meaning that once it accepts a number of clients, it will only be able to then forcibly disconnect those clients without allowing any of them to reconnect. Yet again, we acknowledge that this is not a realistic scenario given the nature of distributed systems, in which FL is supposed to be implemented, which are prone to unintended disconnections [35].

It follows that if an adversary is able to reconnect, they are not required to stick with the same attack approach they used throughout their previous connection. It is more realistic to assume that an adversary will notice the ineffectiveness of their strategy and may attempt to utilise a different approach once reconnected. This evident gap motivates us to formalise this threat, and provide a tangible definition for its occurrence.

Definition 1 (Reconnecting Malicious Clients). *Given an arbitrary set of $k \in \mathbb{N}$ attack strategies $S \in \{S_1, S_2, \dots, S_k\}$, if an adversary attempts a detectable attack strategy S_i within a dropout-tolerant, reconnection-efficient Federated Learning environment (which they are forcibly disconnected from), upon reconnection, an adversary may switch their attempted attack strategy to S_j such that $j \neq i$ with intentions of bypassing the defence mechanisms strategy S_i was insufficient against.*

As such, besides surveying effectiveness of current attack methods against baseline FL defence mechanisms, our main aim is to provide evidence for Reconnecting Malicious Clients as a form of adaptive adversary. To the best of our knowledge, our work is going to be the first to showcase this kind of evidence for such concerns.

3.6.2. Adaptive Attacks

To highlight the fundamental differences that adaptive adversaries are characterised with and further continue our discussion, we will introduce the notion of adaptive attacks that refer back to the previous notions known throughout literature, while simultaneously incorporating our definition of RMCs. We first inspire ourselves with the notion presented in [113], where authors showcase a constrain-and-scale method with the following expression.

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{class} + (1 - \alpha) \mathcal{L}_{anomaly} \quad (14)$$

Where the point is to introduce another loss objective for the anomaly detection $\mathcal{L}_{anomaly}$, while sustaining the loss capture across every class $\mathcal{L}_{class}$. Controlled by the parameter $\alpha \in [0, 1]$ that

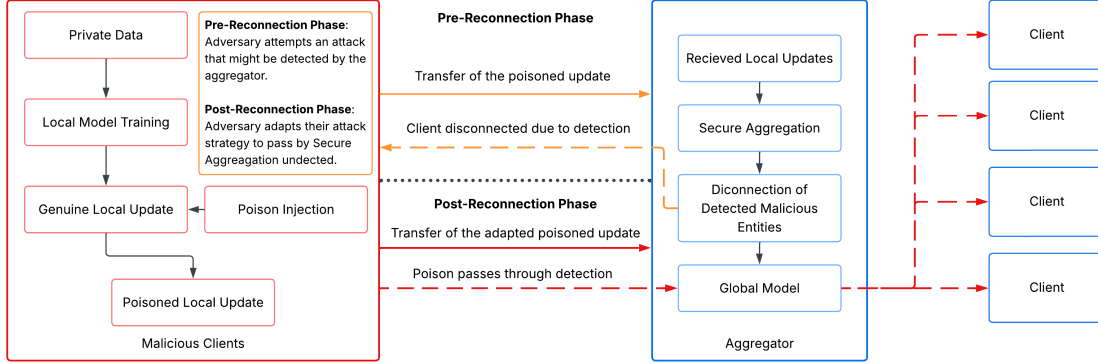


Figure 5: Visualised concept of Reconnecting Malicious Clients (RMCs)

balances between the severity of the attack and it's adaptation against the defence mechanisms. Intuitively, this will result in a forced compromise an adversary has to take which is known throughout literature as an adversarial dilemma [114]. Constrain-and-scale attack has shown to serve throughout literature as the traditional approach to adaptive attacks, with every extension of it being reducible to the authors proposed expression. A good example may be found in [40] where the authors also take from [113] and incorporate a harder to detect approach by choosing an initially low α value that is gradually increased over time to optimise the severity of the attack while maintaining the undetectability. Similarly, any approach that is simpler yet still incorporates a hyper-parameter that optimises the attack effectiveness and stealth may be extended to that expression.

With that said, we broaden the aforementioned method to encapsulate Reconnecting Malicious Clients in the following manner.

Definition 2 (Adaptive Attack). *Given an arbitrary set of $k \in \mathbb{N}$ strategies $S \in \{S_1, S_2, \dots, S_k\}$, where each strategy S_i is reducible or extendible to constrain-and-scale method with a loss objective $\mathcal{L}_{anomaly}$ and a balancing hyper-parameter $\alpha \in [0, 1]$. An attack is called adaptive when an adversary may arbitrarily adapt their attack to a target secure aggregation protocol by modifying α , switching strategy to S_j such that $j \neq i$, or both.*

What follows from our definition is that we may extend the previously established constrain-and-scale method with the following expression:

$$\mathcal{L}_{\mathcal{A}(S, \alpha)} = S_i[\alpha \mathcal{L}_{class} + (1 - \alpha) \mathcal{L}_{anomaly}] \quad \text{where} \quad S = \{S_1, \dots, S_k : k \in \mathbb{N}\}, \quad \alpha \in [0, 1] \quad (15)$$

Where simply S is a set of all strategies an adversary $\mathcal{A}$ is capable of executing with an arbitrary α . Intuitively, this allows us for further extension towards adversarial optimisation in order to establish an optimal strategy S_i that achieves maximum severity without detection among all strategies an adversary considers implementing. This optimisation problem may be defined in the following way:

$$\mathcal{A}^*(S, \alpha) = \arg \max_i (S_i) \quad \text{where} \quad \mathcal{L}_{\mathcal{A}(S_i, \alpha)} = \max_{\mathcal{L}_{class}} \{S_1, S_2, \dots, S_k\} \quad (16)$$

Where again, all strategies S_i are reducible to constrain-and-scale method, and pass through undetected by the FL detection systems and defences. This way we have managed to expand the current definition of adaptive attacks as well as adversaries, with respect to the recent literature.

4. Methodology

To highlight the state of current baseline methods of defence, in this section we conduct experiments which will showcase the following: (1) we produce results using selected baseline attack vectors on an unsecured FL system, (2) we then showcase the effect of implemented defence mechanisms as well as their configuration in presence of selected attacks, and (3), we introduce our novel method of attack to the FL system with already existing defence mechanisms, describe our generalizations and present their effectiveness.

4.1. Motivation for Chosen Attacks and Defences

The selection of specific attacks and defences in this study is driven by a balance between relevance to contemporary literature and accessibility for readers less familiar with FL security. The three attack models employed (Gaussian Noise, ALIE, and IPM) are well-represented in state-of-the-art FL security research and encapsulate a spectrum of threat characteristics, from stochastic perturbations to more structured and targeted manipulations. These attack strategies collectively capture essential paradigms of model poisoning, thereby providing a representative foundation for exploring adversarial dynamics in FL systems.

Similarly, the choice of defence mechanisms (Krum and Trimmed Mean) reflect the conceptual clarity of secure aggregation strategies in the literature. Both algorithms are widely cited and serve as canonical examples of robust aggregation rules designed to mitigate Byzantine approaches against FL systems. Crucially, their algorithmic simplicity ensures that the mechanics can be readily understood without requiring an extensive background in cryptographic or statistical learning theory, making them suitable for inclusion in a survey intended to be broadly accessible.

While this paper surveys a broader array of attacks and defensive strategies, the implementation of these selected techniques is deemed sufficient for the Proof of Concept demonstrations central to our work. Specifically, these methods provide an appropriate baseline for analysing the impact of the newly identified Reconnecting Malicious Clients approach for adaptive adversaries. By focusing on widely adopted and clear mechanisms, the experimental design ensures that observed effects can be attributed with confidence to the dynamics introduced by RMCs, rather than confounding complexities of highly specialised techniques.

4.2. Deployment

To provide clarification on the process we have conducted our experiments with, this segment will elaborate on deployment of the experiments.

We deploy all 40 clients within a virtual environment along with a single aggregator in a centralised HFL setting, where clients get isolated access to their fraction of the data from BRFSS dataset. Then, 19 clients are sampled to behave maliciously as per the $\lceil \frac{N}{2} \rceil - 1$ assumption, and have their attack parameters configured. This setup is then used to conduct four rounds of experiments: (1) establishing the base performance where secure aggregation protocols are absent, (2) testing the performance of Trimmed Mean protocol against all attacks, (3) testing the performance of Krum against the same attacks, and (4), implementing Reconnecting Malicious Clients that switch from Gaussian Noise attack to ALIE attack. For the purposes of comparison, reconnections are simulated such that all malicious clients begin to change their attack strategy at around round 30, which is equivalent to their forcible disconnections at around the same time. Following both of our definitions, such adaptive attacks are free to change strategies they have been previously attempting to degrade the global model with. Lastly, we then record the accuracy of all experiment rounds for attacks against aggregation protocols, and store them for further analysis.

5. Results and Discussion

5.1. Experimental Setup

As previously mentioned, our study conducts a simulation of a centralised HFL environment that has an aggregator with 40 clients. The task chosen for the FL system to undertake was diabetes prediction (binary classification), given the relevance of ML methods used in healthcare. Specifically, we use a cleaned dataset from [115], which consists of 70692 survey responses from the

Behavioural Risk Factor Surveillance System (BRFSS 2015). We distribute data evenly across all 40 clients for simplicity; each client will separate its own data into both training and testing data, with 30% of data being used for testing, with overall accuracy being computed through the average accuracy of the received global model across all clients.

For our global model, this work uses a simple feedforward neural network with $8 * 16 * 8$ fully connected layers with applied learning rate of $\alpha = 0.01$ and a dropout $p = 0.5$ between all layers until the last two, where the dropout is set to be at $p = 0.2$ between last hidden layer and the output layer. We set the batch size of our training and testing loaders to 32 and we ensure to shuffle the training data prior to training. For simplicity, we imitate FedSGD for our baseline aggregation algorithm where we compute a simple average of the gradients as we distribute the data equally among all clients, averaging across all gradients.

Lastly, we highlight that the attack vectors used for our experiments are ALIE, IPM and Gaussian-Noise attacks as our baseline attacks for evaluation. Primarily because we regard those attacks as simple for an adversary to implement, as such, they do not require a sophisticated adversary to be carried out, making secure aggregation algorithms much more likely to face those attacks. We argue this choice as well by the fact that we want to show the impact of reconnecting malicious clients and the severity of such adaptive methods of attack even when we assume that the sophistication of the attack itself is not great. At the same time, for our baseline secure aggregation algorithms we use Trimmed Mean and Krum, as they are commonly used throughout the literature to compare novel aggregation algorithms marking them as a baseline within the domain for statistic-based defences. Finally, we assume that the FL system even when in presence of an adversary, still mostly consists of benign clients, as such adversaries control at most $\lceil \frac{N}{2} \rceil - 1$ byzantine clients, serving as our worst-case scenario.

5.2. Attacks Evaluation (Base case)

To present the base case for our evaluation, we showcase the baseline performance of our model under a FedSGD algorithm without any attacks, as visible on the first plot of Figure 4. With no attack present, the global model achieves a satisfiable benchmark performance of 75%. However, once the attacks are introduced to the system the performance drastically degrades down to 50% which makes it no better than a random guess. This appears as no surprise due to the presence of 19 malicious clients as given by $\lceil \frac{N}{2} \rceil - 1$ with 40 total clients, visualising our worst-case scenario.

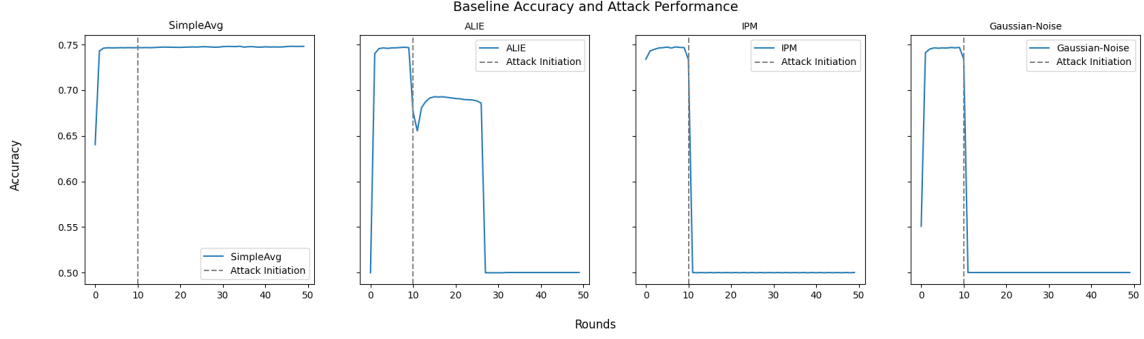


Figure 6: Baseline aggregation algorithm using FedSGD shows the desired performance of the model. When introducing ALIE, Krum and Gaussian-Noise attacks, they show to be effectively degrading performance of the global model after initiation at round 10.

Both Gaussian-Noise and IPM have caused a significant decrease in performance immediately after being introduced to the system. We have used $\epsilon = 15$ for the IPM attack and set the μ and σ to 0.5 for our Gaussian noise, which in the current system presents a significant disturbance of parameters, explaining the rapid degradation in performance. Those attacks also do not attempt to adapt themselves to the rest of the clients which maximises severity at a faster rate when no defences are present.

ALIE was the only attack to behave differently, as it also reached the maximum severity by dropping global model accuracy down to 50%. However, after attack initiation it has first dropped down to 65-69% range, and only after 26th round, it further degraded down to 50%. This is a known effect that ALIE can have on models [93] which is most likely due to the fact that the injected poisons to the local models that are being uploaded by the byzantine clients are bounded to the models performance of benign clients, limiting the immediate severity of the attack yet achieving its maximum potential a while later. We find that the z^{max} value for our attack is set to be approximately 1.3 as it follows from [99] using $\max_z(\phi(z) < \frac{n-m-s}{n-m})$. Given 40 clients with 19 malicious workers and 2 support workers, our value satisfies $\phi(z) < \frac{19}{21}$ and is reasonably close to the maximum.

5.3. Defences Evaluation

To benchmark the security measures used, we applied Krum and Trimmed Mean. Given the best-case scenario, the FL system may be prepared for in terms of determining aggregation

parameters. That is, for Trimmed Mean, we used $\beta = 0.4$, assuming that the FL system may afford to trim 80% of its clients, ensuring a high degree of security and keeping strictly to a narrow fraction of clients around the mean for aggregation. Whereas for Krum, we have set the expected value for byzantine clients the aggregator will operate on to $\lceil \frac{N}{2} \rceil - 1$, again, assuming that the FL system can afford to lose around half of its clients in the aggregation process. These strong security notions are here to highlight the potential severity of the baseline attacks presented.

5.3.1. *Trimmed Mean*

Figure 5 presents the results of our experiments conducted on the Trimmed Mean. In our case, the worst-performing attack was IPM, which is most likely caused by the fact that manipulating the inner product to be negative will face poisoned vectors opposite to the general direction that benign vectors are facing. As such, with our assumption stating that the majority of clients are benign, the resulting poisoned vectors are inevitably going to appear on both ends of the distribution around the mean and therefore trimmed by the aggregator, causing no noticeable damage to the global model. This is going to be the general result for most attacks that are based around manipulating the direction of vectors in such a way that the exact position within the distribution cannot be controlled.

Gaussian-Noise followed a similar pattern; however, due to the randomness introduced by the attack based on its distribution, it most likely involved a small fraction of Byzantine clients in the aggregation. Causing fluctuations in the global models' performance, however, the damage caused appears to be relatively minimal. We note that in case of Noise attacks put on Trimmed-Mean, swaying poisoned parameters to be strictly positive or negative may increase the effectiveness of the attack, as the trimmed mean doesn't differentiate between the two tails of the distribution for the updates it receives. As such, large-scale attacks that follow our assumption and take up almost half of the total number of clients still show some effectiveness, as all of the Byzantine clients are located on one side of the distribution. Where some still are able to find themselves within the range that the Trimmed Mean aggregates around. Given that we have used values of 0.5 for both μ and σ , this is also a likely explanation for the fluctuations.

ALIE was most effective as expected, with its poison bounds, ALIE is kept revolving around the distribution of clients while still disrupting their performance. In our experiments, it has managed to decrease global models' performance below a 70% threshold, managing to reach as low

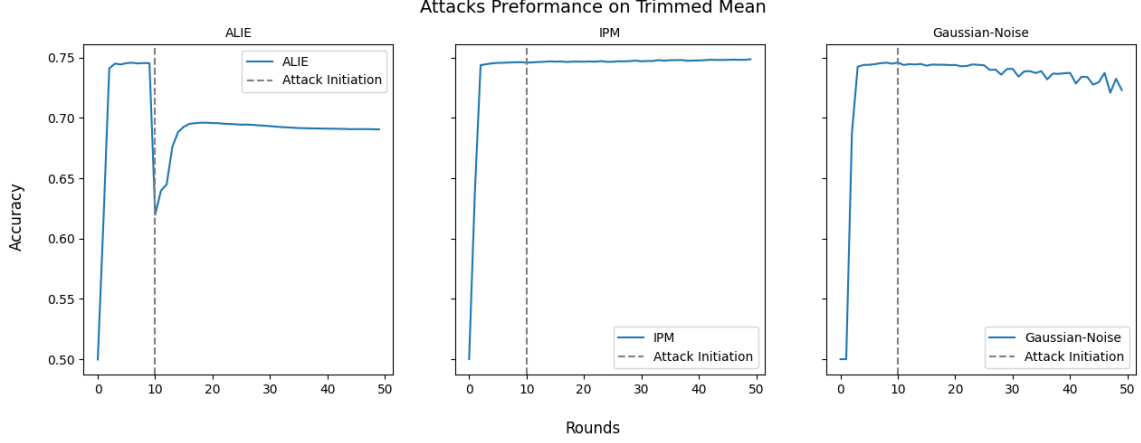


Figure 7: Attack performed on Trimmed Mean presenting the highest effectiveness of methods that concentrate around adapting to the aggregate parameters.

as 62%. Showing substantial effectiveness against defensive measures that use the distribution of all clients for aggregation. In general, attack methods that adapt or will aim to optimise for a specific aggregate parameter have high potential, as the incorporated optimisation problems are adaptive by nature [116]. Meaning that as long as the attack is not detected, it may continue to optimise for aggregate parameters and most often significantly decrease the performance of global models.

5.3.2. Krum

As previously described, Krum introduces a different approach to defences [104] which concentrates on the distances between vectors received for aggregation. Our experimental results, as visible on Figure 6, provide substantial evidence that this approach is more effective against all of our baseline attacks. To be specific, introducing score mechanisms based on an arbitrary metric like distances allows for the aggregation mechanism to move away from methods that generalise the population. Choosing a specific update that is deemed as most reliable for a global model parameter shows to be more secure.

This highlights the significance that the score-based aggregation plays in the FL domain. As many state-of-the-art aggregation methods like FLTrust [101] or FLARE [117] show, they utilise some arbitrary scoring systems that allow the aggregator to establish the most reliable update that it can base its computations on.

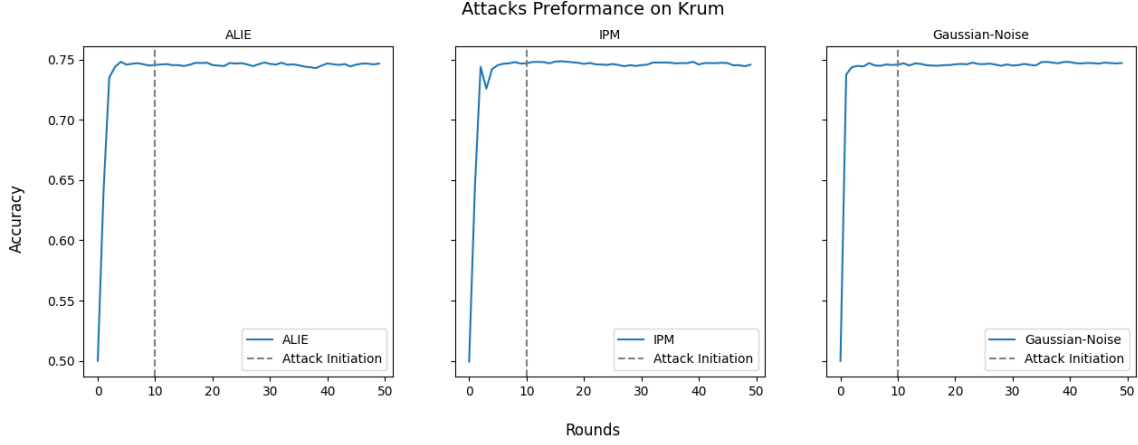


Figure 8: Attacks that attempted to target Krum aggregation rule have failed to cause any substantial damage to the global models performance under our strong security assumptions, highlighting the effectiveness of score-based approaches.

5.3.3. Reconnecting Malicious Clients (Adaptive Adversary)

We now bring the essence of this survey, which is to show evidence that adaptive adversaries, which in our scenario are capable of modifying their attack strategy after reconnecting and are capable of degrading global models' performance. The problem we are targeting in this survey is the fact that current defences in FL security, including the most effective methods, do not account for adversaries adapting their attack even after being filtered out by the aggregation. In most practical scenarios, the presence of active adversarial clients is undesirable and thus are disconnected from the system after detection. However, modern FL systems are in need of reconnection efficiency to handle benign clients that unexpectedly disconnect from the system [35]. Current research concentrates on dropout-tolerance and ways to enhance the performance of the global models using disconnected clients [118, 119, 120]. As such, FL systems allow for reconnections of clients throughout training rounds of the global model, which serves as an attack vector for adversaries.

Adversaries may utilise the systems' ability to efficiently accept reconnecting clients if those systems do not strictly keep track of the disconnected parties that have been rejected, meanwhile accepting new clients for training once the training starts. This is a reasonable scenario to assume, as accepting new clients in FL systems often is said to effectively address problems with collective computation of the system as well as data quantity to prevent overfitting [121].

In this experiment, we simulate a scenario where the FL system actively accepts new clients for training and uses a byzantine-robust aggregation rule before computing the global model using the same aggregate parameters as before. We allow for the adversary to first attempt a Gaussian-Noise attack with the same configurations as in previous experiments, but this time, the system detects and forcibly disconnects Byzantine clients from aggregation. The adversary then reconnects to the system, but this time adapts from the previous disconnection and switches the attack method to ALIE again with $z^{max} = 1.3$, using ALIE attack to utilise support vectors as it requires that they need to be statistically closer to the benign vectors. This effectively simulates a scenario where Byzantine clients controlled by an adaptive adversary are forcibly disconnected after getting detected, and then the adversary proceeds to adapt their attack and reconnect to the system.

Figure 7 presents our results on both Krum and Trimmed Mean aggregation rules, both of which falter to this method of attack. Trimmed Mean has completely failed to stop the attack, resulting in a performance that is no better than a random guess. It is once again due to the approach ALIE takes to bound the poison in accordance with the population, but this time shows more severe impact most likely due to the fact that the adversary in such scenarios can manipulate the value of z^{max} to adapt variance inflation it introduces such that more byzantine clients will be included in the aggregation.

Similarly, Krum has suffered a noticeable impact of the adapted attack, as the adapted z^{max} value exploits Krum’s heavy reliance on distance metrics. Mimicking the statistical distribution of benign to be much closer than what was shown before caused Krum to involve Byzantine clients in the aggregation, and although it limits the impact of the attack, it is still managing to break the aggregation defences. There is a potential that this could be exploited in other aggregation rules that introduce score-based systems or reputation mechanisms, as an adversary can repeatedly attempt to optimise for the maximum severity and minimum detectability of Byzantine clients. That is only if the FL system cannot retain knowledge about disconnected clients and the reasons for their disconnection.

We note that this type of adaptive adversarial attack cannot be left unaddressed, as it gives adversaries too great a potential for damage they can cause to various FL systems and their global models. For the reader’s convenience, our results are summarised in Table 1.

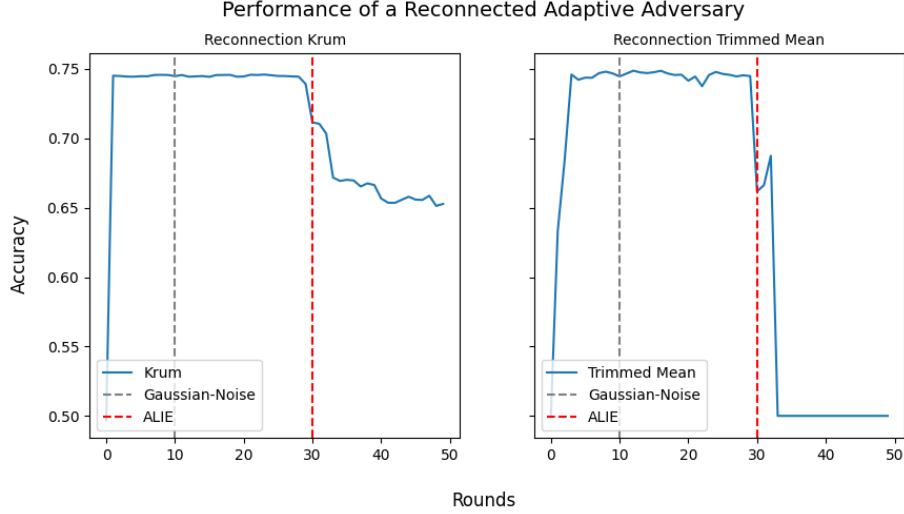


Figure 9: Adaptive adversaries using clients that are not recognised again as malicious effectively degrade models’ performance, highlighting the potential severity of this form of attack.

Table 1: Experimental results for baseline attacks against Trimmed Mean and Krum secure aggregation algorithms, compared to the reconnecting malicious clients’ method of attack.

Algorithm	No Attack	Gaussian	IPM	ALIE	Reconnecting Clients	
					Gaussian	ALIE
No Defence	0.748	0.500	0.500	0.500		
Trimmed Mean		0.723	0.748	0.691	0.744	0.500
Krum		0.747	0.745	0.746	0.745	0.653

5.4. Statistical Validity

To evaluate the statistical significance of RMCs on Secure Aggregation protocols, we conducted paired t-tests on two separate experimental setups for the two chosen algorithms. The intent was to assess whether the simulated reconnection of previously disconnected malicious clients significantly altered the system’s performance. To reiterate, each experiment was structured around two phases: an initial attack phase (rounds 10–29) and a reconnection-based attack (rounds 30–50).

A paired two-tailed t-test was applied to both datasets (corresponding to each aggregation algorithm), comparing the mean accuracies between the two attack phases. The null hypothesis H_0 assumes no significant difference between the two means (i.e., the reconnection has no effect), while

our alternative hypothesis H_a posits that the means differ significantly due to the reconnection strategy posed by RMCs. All statistical evaluations were conducted at a 99.73% confidence level, equivalent to a 3σ threshold. The use of this stricter criterion was motivated by the need to minimize Type I errors and ensure that observed differences are not attributable to random variance.

For both algorithms, the statistical analysis strongly supports the alternative hypothesis, indicating a significant impact of RMCs behaviour post-reconnection. When performed on the results from Krum aggregation rule, the validity tests produced a t-score of 18.29 with a confidence interval standing at $[0.686, 0.727]$ where Krum’s accuracy mean during the initial attack phase (0.745) lied outside. Similarly, the tests from Trimmed Mean results produced a t-score of 15.57 along with a confidence interval $[0.576, 0.696]$ with the initial attack mean of 0.744.

Across both experimental setups and algorithms, the paired t-test results affirm that the reconnection of malicious clients introduces a statistically significant degradation in system performance. The t-scores and corresponding p-values exceed conventional significance thresholds even under a strict 99.73% confidence level. Furthermore, the exclusion of the first attack mean from the second attack’s confidence interval substantiates that the observed effects are unlikely to be artefacts of noise or natural system variability. These findings highlight the critical threat posed by RMCs and underscore the need for secure aggregation protocols to incorporate explicit mechanisms for mitigating reconnection-based adversarial strategies.

6. Research Directions

Having surveyed the current state-of-the-art for federated learning security, with specific emphasis on robustness of model aggregation. We now proceed to derive directions in which future research should be conducted based on the lessons learned from this survey.

Adaptive Adversaries: Definitions. As previously stated, security in FL suffers from a lack of widely accepted and formal definitions of adaptive adversaries, with the first notable attempts visible in [113]. This directly results in our inability to effectively categorise their impact along with methods of attack, reverberating the challenges with optimisation and generalisation of our defence methods against them. Further, also concerns our current state-of-the-art security mechanisms, where it is not possible to evaluate their effectiveness against adaptive adversaries without having properly defined them. Although we have put forward an attempt to define adaptive attacks using Definition 2 that incorporates the potential use of Reconnecting Malicious Clients,

further exploration is required to test and solidify such definitions as our understanding of adaptive adversaries broadens.

Adaptive Adversaries: Threats. This survey presented one of the potential approaches an adaptive adversary may take for their attack. However, adaptive adversaries have multiple points of entry for their approaches besides reconnecting to the actively trained FL system. That being said, it still stands that the most common approach seen throughout surveyed literature is to allow an adversary to incorporate a hyperparameter that maximises severity while maintaining the stealth of their attack, essentially operating on some form of an optimisation problem. [122] A more sophisticated example of this may be found in [40] where authors further developed adversarial adaptation by first extracting defence metrics using benign training and then use augmented Lagrange optimisation to reduce deviations to a valid range that would be allowed by the defence mechanisms.

With an emerging threat that RMCs pose adding to the pile of potential approaches such an adversary may take in order to attack global models, it only broadens the plethora of possible threats posed by adaptive adversaries and their impact on FL environments.

FL in Healthcare. This survey used a diabetes dataset to also prove that the question of adaptive adversaries especially applies to healthcare-related tasks and urges the community to revisit this threat. Acknowledging the fact that domains such as healthcare, medicine and autonomous vehicles often can't afford to experience even the slightest disturbances in performance, given the sensitivity of their tasks, attacks do not necessarily need to aim for maximised severity in terms of the global model's performance. Instead, they can optimise for the undetectability of Byzantine clients and still achieve a satisfiable impact on the model, rendering it unusable in such contexts. We believe that using the presented evidence in this survey, this work can bring attention to this threat in relation to those domains.

Further Experiments. We acknowledge that there is a need for further exploration of adaptive adversaries in different scenarios. Here, we have only evaluated the effects of reconnecting malicious clients on an FL system that actively accepts new clients during training. Although not unreasonable, this does not cover all highly possible scenarios in which this method of attack could be used in. Example being systems that do not follow centralised FL architecture like we did in our experiments, but use decentralised FL. Hence, we call for further experiments from the research community to collectively better understand this threat.

Solutions Over Communication Channels. As a final remark of this survey, we bring attention to the potential solutions this threat may be countered with. To be concrete, it necessitates that RMCs are tracked to pose difficulties for the adaptive adversary to reconnect unnoticed by the system. To our aid may come methods used over communication channels that would cryptographically verify all clients during training when receiving updates and distributing global models, such that it is capable of recognising previous malicious clients that should not be accepted back into the system. The least this could achieve is a significant increase in the required sophistication of the attack an adversary attempts to carry out to achieve the same level of effectiveness.

7. Conclusions

Ensuring model robustness in federated learning proves to be one of the leading security challenges in this domain. In order to preserve the integrity of global models, the research community has considered a variety of statistics-based methods through aggregation to detect, reject and disconnect malicious clients. However, it was noticed that current research does not consider adaptive adversaries, which may optimise their attack such that it will minimise detectability at maximised severity.

This survey reviews current baseline secure aggregation methods and attack vectors, showcasing their effectiveness under strong FL security assumptions for their aggregate parameters. Then, we present an unexplored method of attack involving reconnecting malicious clients and show evidence for their effectiveness as well as relate them back to the concept of adaptive adversaries. We bring further attention to the lack of research in this direction as well as the lack of definitions, highlighting the need for further explorations around the threat of adaptive adversaries, as well as providing the first baseline evidence for this concern, which we hope will bring more eyes to this unexplored problem.

8. Data share statement

Data will be made available on request.

9. CRediT authorship contribution statement

Jakub Kacper Szelaġ: Conceptualisation, Writing - original draft, Writing - review and editing, Data curation, Formal analysis, Investigation, Methodology, Visualisation, Validation, Software. **Ji-Jian Chin:** Conceptualisation, Project administration, Supervision, Writing - review and editing. **Sook-Chin Yip:** Project administration, Writing - review and editing, Funding acquisition.

10. Acknowledgement

This research was supported by the Ministry of Higher Education Malaysia through the Fundamental Research Grant Scheme (FRGS/1/2023/ICT07/MMU/03/1).

References

- [1] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, pages 1273–1282. PMLR, April 2017. URL <https://proceedings.mlr.press/v54/mcmahan17a.html>. ISSN: 2640-3498.
- [2] Chen Zhang, Yu Xie, Hang Bai, Bin Yu, Weihong Li, and Yuan Gao. A survey on federated learning. *Knowledge-Based Systems*, 216:106775, March 2021. ISSN 0950-7051. doi: 10.1016/j.knosys.2021.106775. URL <https://www.sciencedirect.com/science/article/pii/S0950705121000381>.
- [3] Li Li, Yuxi Fan, Mike Tse, and Kuo-Yi Lin. A review of applications in federated learning. *Computers & Industrial Engineering*, 149:106854, November 2020. ISSN 0360-8352. doi: 10.1016/j.cie.2020.106854. URL <https://www.sciencedirect.com/science/article/pii/S0360835220305532>.
- [4] Hao Li, Chengcheng Li, Jian Wang, Aimin Yang, Zezhong Ma, Zunqian Zhang, and Dianbo Hua. Review on security of federated learning and its application in healthcare. *Future Generation Computer Systems*, 144:271–290, July 2023. ISSN 0167-739X. doi:

10.1016/j.future.2023.02.021. URL <https://www.sciencedirect.com/science/article/pii/S0167739X23000626>.

- [5] Lingjuan Lyu, Han Yu, Xingjun Ma, Chen Chen, Lichao Sun, Jun Zhao, Qiang Yang, and Philip S. Yu. Privacy and Robustness in Federated Learning: Attacks and Defenses. *IEEE Transactions on Neural Networks and Learning Systems*, 35(7):8726–8746, July 2024. ISSN 2162-2388. doi: 10.1109/TNNLS.2022.3216981. URL <https://ieeexplore.ieee.org/document/9945997/?arnumber=9945997>. Conference Name: IEEE Transactions on Neural Networks and Learning Systems.
- [6] Kai Hu, Sheng Gong, Qi Zhang, Chaowen Seng, Min Xia, and Shanshan Jiang. An overview of implementing security and privacy in federated learning. *Artificial Intelligence Review*, 57(8):204, July 2024. ISSN 1573-7462. doi: 10.1007/s10462-024-10846-8. URL <https://doi.org/10.1007/s10462-024-10846-8>.
- [7] Virraji Mothukuri, Reza M. Parizi, Seyedamin Pouriyeh, Yan Huang, Ali Dehghantanha, and Gautam Srivastava. A survey on security and privacy of federated learning. *Future Generation Computer Systems*, 115:619–640, February 2021. ISSN 0167-739X. doi: 10.1016/j.future.2020.10.007. URL <https://www.sciencedirect.com/science/article/pii/S0167739X20329848>.
- [8] Nader Bouacida and Prasant Mohapatra. Vulnerabilities in Federated Learning. *IEEE Access*, 9:63229–63249, 2021. ISSN 2169-3536. doi: 10.1109/ACCESS.2021.3075203. URL <https://ieeexplore.ieee.org/document/9411833/?arnumber=9411833>. Conference Name: IEEE Access.
- [9] Lingjuan Lyu, Han Yu, and Qiang Yang. Threats to Federated Learning: A Survey, March 2020. URL <http://arxiv.org/abs/2003.02133>. arXiv:2003.02133 [cs].
- [10] Akarsh K. Nair, Ebin Deni Raj, and Jayakrushna Sahoo. A robust analysis of adversarial attacks on federated learning environments. *Computer Standards & Interfaces*, 86: 103723, August 2023. ISSN 0920-5489. doi: 10.1016/j.csi.2023.103723. URL <https://www.sciencedirect.com/science/article/pii/S0920548923000041>.
- [11] Jiale Zhang, Junjun Chen, Di Wu, Bing Chen, and Shui Yu. Poisoning Attack in Federated Learning using Generative Adversarial Nets. In *2019 18th IEEE International Conference On*

- Trust, Security And Privacy In Computing And Communications/13th IEEE International Conference On Big Data Science And Engineering (TrustCom/BigDataSE)*, pages 374–380, August 2019. doi: 10.1109/TrustCom/BigDataSE.2019.00057. URL <https://ieeexplore.ieee.org/document/8887357/>. ISSN: 2324-9013.
- [12] Ibrahim M. Ahmed and Manar Younis Kashmoola. Investigated Insider and Outsider Attacks on the Federated Learning Systems. In *2022 IEEE International Conference on Communication, Networks and Satellite (COMNETSAT)*, pages 438–443, November 2022. doi: 10.1109/COMNETSAT56033.2022.9994330. URL <https://ieeexplore.ieee.org/document/9994330/?arnumber=9994330>.
- [13] Nuria Rodríguez-Barroso, Daniel Jiménez-López, M. Victoria Luzón, Francisco Herrera, and Eugenio Martínez-Cámara. Survey on federated learning threats: Concepts, taxonomy on attacks and defences, experimental study and challenges. *Information Fusion*, 90:148–173, February 2023. ISSN 1566-2535. doi: 10.1016/j.inffus.2022.09.011. URL <https://www.sciencedirect.com/science/article/pii/S1566253522001439>.
- [14] Minghong Fang, Zifan Zhang, Hairi, Prashant Khanduri, Jia Liu, Songtao Lu, Yuchen Liu, and Neil Gong. Byzantine-Robust Decentralized Federated Learning. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security, CCS '24*, pages 2874–2888, New York, NY, USA, December 2024. Association for Computing Machinery. ISBN 979-8-4007-0636-3. doi: 10.1145/3658644.3670307. URL <https://dl.acm.org/doi/10.1145/3658644.3670307>.
- [15] Yupeng Jiang, Yong Li, Yipeng Zhou, and Xi Zheng. Sybil Attacks and Defense on Differential Privacy based Federated Learning. In *2021 IEEE 20th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, pages 355–362, October 2021. doi: 10.1109/TrustCom53373.2021.00062. URL <https://ieeexplore.ieee.org/abstract/document/9724328>. ISSN: 2324-9013.
- [16] Thomas Werthenbach and Johan Pouwelse. Towards Sybil Resilience in Decentralized Learning, June 2023. URL <http://arxiv.org/abs/2306.15044>. arXiv:2306.15044 [cs].
- [17] Kaiyue Zhang, Xuan Song, Chenhan Zhang, and Shui Yu. Challenges and future directions of secure federated learning: a survey. *Frontiers of Computer Science*, 16(5):165817, December

2021. ISSN 2095-2236. doi: 10.1007/s11704-021-0598-z. URL <https://doi.org/10.1007/s11704-021-0598-z>.

- [18] Helio N. Cunha Neto, Jernej Hribar, Ivana Dusparic, Diogo Menezes Ferrazani Mattos, and Natalia C. Fernandes. A Survey on Securing Federated Learning: Analysis of Applications, Attacks, Challenges, and Trends. *IEEE Access*, 11:41928–41953, 2023. ISSN 2169-3536. doi: 10.1109/ACCESS.2023.3269980. URL <https://ieeexplore.ieee.org/abstract/document/10107622>. Conference Name: IEEE Access.
- [19] Rémi Gosselin, Loïc Vieu, Faiza Loukil, and Alexandre Benoit. Privacy and Security in Federated Learning: A Survey. *Applied Sciences*, 12(19):9901, January 2022. ISSN 2076-3417. doi: 10.3390/app12199901. URL <https://www.mdpi.com/2076-3417/12/19/9901>. Number: 19 Publisher: Multidisciplinary Digital Publishing Institute.
- [20] Krishna Pillutla, Sham M. Kakade, and Zaid Harchaoui. Robust Aggregation for Federated Learning. *IEEE Transactions on Signal Processing*, 70:1142–1154, 2022. ISSN 1941-0476. doi: 10.1109/TSP.2022.3153135. URL <https://ieeexplore.ieee.org/abstract/document/9721118>. Conference Name: IEEE Transactions on Signal Processing.
- [21] Syed Zawad, Ahsan Ali, Pin-Yu Chen, Ali Anwar, Yi Zhou, Nathalie Baracaldo, Yuan Tian, and Feng Yan. Curse or Redemption? How Data Heterogeneity Affects the Robustness of Federated Learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(12):10807–10814, May 2021. ISSN 2374-3468. doi: 10.1609/aaai.v35i12.17291. URL <https://ojs.aaai.org/index.php/AAAI/article/view/17291>. Number: 12.
- [22] Jingxue Chen, Hang Yan, Zhiyuan Liu, Min Zhang, Hu Xiong, and Shui Yu. When Federated Learning Meets Privacy-Preserving Computation. *ACM Comput. Surv.*, 56(12):319:1–319:36, October 2024. ISSN 0360-0300. doi: 10.1145/3679013. URL <https://doi.org/10.1145/3679013>.
- [23] Xuefei Yin, Yanming Zhu, and Jiankun Hu. A Comprehensive Survey of Privacy-preserving Federated Learning: A Taxonomy, Review, and Future Directions. *ACM Comput. Surv.*, 54(6):131:1–131:36, July 2021. ISSN 0360-0300. doi: 10.1145/3460427. URL <https://dl.acm.org/doi/10.1145/3460427>.

- [24] Stacey Truex, Nathalie Baracaldo, Ali Anwar, Thomas Steinke, Heiko Ludwig, Rui Zhang, and Yi Zhou. A Hybrid Approach to Privacy-Preserving Federated Learning. In *Proceedings of the 12th ACM Workshop on Artificial Intelligence and Security*, AISec'19, pages 1–11, New York, NY, USA, November 2019. Association for Computing Machinery. ISBN 978-1-4503-6833-9. doi: 10.1145/3338501.3357370. URL <https://doi.org/10.1145/3338501.3357370>.
- [25] Tabassum Anika. A systematic literature review on untargeted model poisoning attacks and defense mechanisms in federated learning. *Systematic Literature Review and Meta-Analysis Journal*, 3(4):117–126, 2022. ISSN 2753-9148. doi: 10.54480/slr-m.v3i4.42. URL <https://slr-m.com/index.php/home/article/view/42>. Number: 4.
- [26] Harsh Kasyap and Somanath Tripathy. Beyond data poisoning in federated learning. *Expert Systems with Applications*, 235:121192, January 2024. ISSN 0957-4174. doi: 10.1016/j.eswa.2023.121192. URL <https://www.sciencedirect.com/science/article/pii/S0957417423016949>.
- [27] Zheyi Chen, Pu Tian, Weixian Liao, and Wei Yu. Towards multi-party targeted model poisoning attacks against federated learning systems. *High-Confidence Computing*, 1(1):100002, June 2021. ISSN 2667-2952. doi: 10.1016/j.hcc.2021.100002. URL <https://www.sciencedirect.com/science/article/pii/S2667295221000039>.
- [28] Vale Tolpegin, Stacey Truex, Mehmet Emre Gursoy, and Ling Liu. Data Poisoning Attacks Against Federated Learning Systems. In Liqun Chen, Ninghui Li, Kaitai Liang, and Steve Schneider, editors, *Computer Security – ESORICS 2020*, pages 480–501, Cham, 2020. Springer International Publishing. ISBN 978-3-030-58951-6. doi: 10.1007/978-3-030-58951-6_24.
- [29] Yuwei Sun, Hideya Ochiai, and Jun Sakuma. Attacking-Distance-Aware Attack: Semi-targeted Model Poisoning on Federated Learning. *IEEE Transactions on Artificial Intelligence*, 5(2):925–939, February 2024. ISSN 2691-4581. doi: 10.1109/TAI.2023.3280155. URL <https://ieeexplore.ieee.org/document/10136777/?arnumber=10136777>. Conference Name: IEEE Transactions on Artificial Intelligence.
- [30] Yichen Wan, Youyang Qu, Wei Ni, Yong Xiang, Longxiang Gao, and Ekram Hossain. Data and Model Poisoning Backdoor Attacks on Wireless Federated Learning, and the

- Defense Mechanisms: A Comprehensive Survey. *IEEE Communications Surveys & Tutorials*, 26(3):1861–1897, 2024. ISSN 1553-877X. doi: 10.1109/COMST.2024.3361451. URL <https://ieeexplore.ieee.org/document/10423783/?arnumber=10423783>. Conference Name: IEEE Communications Surveys & Tutorials.
- [31] Thuy Dung Nguyen, Tuan Nguyen, Phi Le Nguyen, Hieu H. Pham, Khoa D. Doan, and Kok-Seng Wong. Backdoor attacks and defenses in federated learning: Survey, challenges and future research directions. *Engineering Applications of Artificial Intelligence*, 127:107166, January 2024. ISSN 0952-1976. doi: 10.1016/j.engappai.2023.107166. URL <https://www.sciencedirect.com/science/article/pii/S0952197623013507>.
- [32] Hongyi Wang, Kartik Sreenivasan, Shashank Rajput, Harit Vishwakarma, Saurabh Agarwal, Jy-yong Sohn, Kangwook Lee, and Dimitris Papailiopoulos. Attack of the Tails: Yes, You Really Can Backdoor Federated Learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 16070–16084. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/b8ffa41d4e492f0fad2f13e29e1762eb-Abstract.html.
- [33] Virat Shejwalkar, Amir Houmansadr, Peter Kairouz, and Daniel Ramage. Back to the Drawing Board: A Critical Evaluation of Poisoning Attacks on Production Federated Learning. In *2022 IEEE Symposium on Security and Privacy (SP)*, pages 1354–1371, May 2022. doi: 10.1109/SP46214.2022.9833647. URL <https://ieeexplore.ieee.org/document/9833647/?arnumber=9833647>. ISSN: 2375-1207.
- [34] Konstantinos Lazaros, Dimitrios E. Koumadorakis, Aristidis G. Vrahatis, and Sotiris Kotsiantis. Federated Learning: Navigating the Landscape of Collaborative Intelligence. *Electronics*, 13(23):4744, January 2024. ISSN 2079-9292. doi: 10.3390/electronics13234744. URL <https://www.mdpi.com/2079-9292/13/23/4744>. Number: 23 Publisher: Multidisciplinary Digital Publishing Institute.
- [35] Heng Ye, Jiqiang Liu, Hao Zhen, Wenbin Jiang, Bin Wang, and Wei Wang. VREFL: Verifiable and Reconnection-Efficient Federated Learning in IoT scenarios. *Journal of Network and Computer Applications*, 207:103486, November 2022. ISSN 1084-8045. doi: 10.

1016/j.jnca.2022.103486. URL <https://www.sciencedirect.com/science/article/pii/S108480452200131X>.

- [36] Muhammad Habib ur Rehman, Khaled Salah, Ernesto Damiani, and Davor Svetinovic. Towards Blockchain-Based Reputation-Aware Federated Learning. In *IEEE INFOCOM 2020 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, pages 183–188, July 2020. doi: 10.1109/INFOCOMWKSHPS50562.2020.9163027. URL <https://ieeexplore.ieee.org/document/9163027/?arnumber=9163027>.
- [37] Zhendong Song, Hongguang Sun, Howard H. Yang, Xijun Wang, Yan Zhang, and Tony Q. S. Quek. Reputation-Based Federated Learning for Secure Wireless Networks. *IEEE Internet of Things Journal*, 9(2):1212–1226, January 2022. ISSN 2327-4662. doi: 10.1109/JIOT.2021.3079104. URL <https://ieeexplore.ieee.org/document/9428537/?arnumber=9428537>. Conference Name: IEEE Internet of Things Journal.
- [38] Jiawen Kang, Zehui Xiong, Dusit Niyato, Yuze Zou, Yang Zhang, and Mohsen Guizani. Reliable Federated Learning for Mobile Networks. *IEEE Wireless Communications*, 27(2):72–80, April 2020. ISSN 1558-0687. doi: 10.1109/MWC.001.1900119. URL <https://ieeexplore.ieee.org/document/8994206/?arnumber=8994206>. Conference Name: IEEE Wireless Communications.
- [39] Nanqing Dong, Zhipeng Wang, Jiahao Sun, Michael Kampffmeyer, William Knottenbelt, and Eric Xing. Defending Against Poisoning Attacks in Federated Learning with Blockchain, March 2024. URL <http://arxiv.org/abs/2307.00543>. arXiv:2307.00543 [cs].
- [40] Torsten Krauß, Jan König, Alexandra Dmitrienko, and Christian Kanzow. Automatic Adversarial Adaption for Stealthy Poisoning Attacks in Federated Learning. In *Proceedings 2024 Network and Distributed System Security Symposium*, San Diego, CA, USA, 2024. Internet Society. ISBN 978-1-891562-93-8. doi: 10.14722/ndss.2024.241366. URL <https://www.ndss-symposium.org/wp-content/uploads/2024-1366-paper.pdf>.
- [41] Asadullah Tariq, Mohamed Adel Serhani, Farag M. Sallabi, Ezedin S. Barka, Tariq Qayyum, Heba M. Khater, and Khaled A. Shuaib. Trustworthy Federated Learning: A Comprehensive Review, Architecture, Key Challenges, and Future Research Prospects. *IEEE Open Journal of the Communications Society*, 5:4920–4998, 2024. ISSN 2644-125X. doi: 10.

- 1109/OJCOMS.2024.3438264. URL <https://ieeexplore.ieee.org/document/10623386/?arnumber=10623386>. Conference Name: IEEE Open Journal of the Communications Society.
- [42] Jie Wen, Zhixia Zhang, Yang Lan, Zhihua Cui, Jianghui Cai, and Wensheng Zhang. A survey on federated learning: challenges and applications. *International Journal of Machine Learning and Cybernetics*, 14(2):513–535, February 2023. ISSN 1868-808X. doi: 10.1007/s13042-022-01647-y. URL <https://doi.org/10.1007/s13042-022-01647-y>.
- [43] M. Victoria Luzón, Nuria Rodríguez-Barroso, Alberto Argente-Garrido, Daniel Jiménez-López, Jose M. Moyano, Javier Del Ser, Weiping Ding, and Francisco Herrera. A Tutorial on Federated Learning from Theory to Practice: Foundations, Software Frameworks, Exemplary Use Cases, and Selected Trends. *IEEE/CAA Journal of Automatica Sinica*, 11(4):824–850, April 2024. ISSN 2329-9274. doi: 10.1109/JAS.2024.124215. URL <https://ieeexplore.ieee.org/document/10475194/?arnumber=10475194>. Conference Name: IEEE/CAA Journal of Automatica Sinica.
- [44] Sawsan Abdulrahman, Hanine Tout, Hakima Ould-Slimane, Azzam Mourad, Chamseddine Talhi, and Mohsen Guizani. A Survey on Federated Learning: The Journey From Centralized to Distributed On-Site Learning and Beyond. *IEEE Internet of Things Journal*, 8(7):5476–5497, April 2021. ISSN 2327-4662. doi: 10.1109/JIOT.2020.3030072. URL <https://ieeexplore.ieee.org/abstract/document/9220780>. Conference Name: IEEE Internet of Things Journal.
- [45] Kunal Chandiramani, Dhruv Garg, and N Maheswari. Performance Analysis of Distributed and Federated Learning Models on Private Data. *Procedia Computer Science*, 165:349–355, January 2019. ISSN 1877-0509. doi: 10.1016/j.procs.2020.01.039. URL <https://www.sciencedirect.com/science/article/pii/S1877050920300478>.
- [46] Christodoulos Pappas, Dimitris Chatzopoulos, Spyros Lalis, and Manolis Vavalis. IPLS: A Framework for Decentralized Federated Learning. In *2021 IFIP Networking Conference (IFIP Networking)*, pages 1–6, June 2021. doi: 10.23919/IFIPNetworking52078.2021.9472790. URL <https://ieeexplore.ieee.org/document/9472790/?arnumber=9472790>. ISSN: 1861-2288.

- [47] Hao Ye, Le Liang, and Geoffrey Ye Li. Decentralized Federated Learning With Unreliable Communications. *IEEE Journal of Selected Topics in Signal Processing*, 16(3):487–500, April 2022. ISSN 1941-0484. doi: 10.1109/JSTSP.2022.3152445. URL <https://ieeexplore.ieee.org/document/9716792/?arnumber=9716792>. Conference Name: IEEE Journal of Selected Topics in Signal Processing.
- [48] Wei Liu, Li Chen, and Wenyi Zhang. Decentralized Federated Learning: Balancing Communication and Computing Costs. *IEEE Transactions on Signal and Information Processing over Networks*, 8:131–143, 2022. ISSN 2373-776X. doi: 10.1109/TSIPN.2022.3151242. URL <https://ieeexplore.ieee.org/document/9713700/?arnumber=9713700>. Conference Name: IEEE Transactions on Signal and Information Processing over Networks.
- [49] Liangqi Yuan, Ziran Wang, Lichao Sun, Philip S. Yu, and Christopher G. Brinton. Decentralized Federated Learning: A Survey and Perspective. *IEEE Internet of Things Journal*, 11(21):34617–34638, November 2024. ISSN 2327-4662. doi: 10.1109/JIOT.2024.3407584. URL <https://ieeexplore.ieee.org/document/10542323/?arnumber=10542323>. Conference Name: IEEE Internet of Things Journal.
- [50] Michal Yemini, Rajarshi Saha, Emre Ozfatura, Deniz Gündüz, and Andrea J. Goldsmith. Semi-Decentralized Federated Learning with Collaborative Relaying. In *2022 IEEE International Symposium on Information Theory (ISIT)*, pages 1471–1476, June 2022. doi: 10.1109/ISIT50566.2022.9834707. URL <https://ieeexplore.ieee.org/document/9834707/?arnumber=9834707>. ISSN: 2157-8117.
- [51] Minghong Fang, Xiaoyu Cao, Jinyuan Jia, and Neil Gong. Local Model Poisoning Attacks to {Byzantine-Robust} Federated Learning. pages 1605–1622, 2020. ISBN 978-1-939133-17-5. URL <https://www.usenix.org/conference/usenixsecurity20/presentation/fang>.
- [52] Jie Ma, Tianyi Zhou, Guodong Long, Jing Jiang, and Chengqi Zhang. Structured Federated Learning through Clustered Additive Modeling. *Advances in Neural Information Processing Systems*, 36:43097–43107, December 2023. URL https://papers.nips.cc/paper_files/paper/2023/hash/8668fdc7b2ddf55a0e235824c66f2eee-Abstract-Conference.html.
- [53] Boyu Fan, Chenrui Wu, Xiang Su, and Pan Hui. FedTSA: A Cluster-Based Two-Stage Aggregation Method for Model-Heterogeneous Federated Learning. In Aleš Leonardis, Elisa

- Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision – ECCV 2024*, pages 370–389, Cham, 2025. Springer Nature Switzerland. ISBN 978-3-031-73010-8. doi: 10.1007/978-3-031-73010-8_22.
- [54] Felix Sattler, Klaus-Robert Müller, and Wojciech Samek. Clustered Federated Learning: Model-Agnostic Distributed Multi-Task Optimization under Privacy Constraints, October 2019. URL <http://arxiv.org/abs/1910.01991>. arXiv:1910.01991 [cs].
- [55] Liangqiong Qu, Yuyin Zhou, Paul Pu Liang, Yingda Xia, Feifei Wang, Ehsan Adeli, Li Fei-Fei, and Daniel Rubin. Rethinking Architecture Design for Tackling Data Heterogeneity in Federated Learning. pages 10061–10071, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Qu_Rethinking_Architecture_Design_for_Tackling_Data_Heterogeneity_in_Federated_Learning_CVPR_2022_paper.html.
- [56] Matias Mendieta, Taojiannan Yang, Pu Wang, Minwoo Lee, Zhengming Ding, and Chen Chen. Local Learning Matters: Rethinking Data Heterogeneity in Federated Learning. pages 8397–8406, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Mendieta_Local_Learning_Matters_Rethinking_Data_Heterogeneity_in_Federated_Learning_CVPR_2022_paper.html.
- [57] Mang Ye, Xiuwen Fang, Bo Du, Pong C. Yuen, and Dacheng Tao. Heterogeneous Federated Learning: State-of-the-art and Research Challenges. *ACM Comput. Surv.*, 56(3):79:1–79:44, October 2023. ISSN 0360-0300. doi: 10.1145/3625558. URL <https://doi.org/10.1145/3625558>.
- [58] Dianqi Liu, Liang Bai, Tianyuan Yu, and Aiming Zhang. Towards Method of Horizontal Federated Learning: A Survey. In *2022 8th International Conference on Big Data and Information Analytics (BigDIA)*, pages 259–266, August 2022. doi: 10.1109/BigDIA56350.2022.9874186. URL <https://ieeexplore.ieee.org/document/9874186/?arnumber=9874186>. ISSN: 2771-6902.
- [59] Yang Liu, Yan Kang, Tianyuan Zou, Yanhong Pu, Yuanqin He, Xiaozhou Ye, Ye Ouyang, Ya-Qin Zhang, and Qiang Yang. Vertical Federated Learning: Concepts, Advances, and Challenges. *IEEE Transactions on Knowledge and Data Engineering*, 36(7):3615–3634, July 2024. ISSN 1558-2191. doi: 10.1109/TKDE.2024.3352628. URL <https://ieeexplore.ieee>.

org/document/10415268/?arnumber=10415268. Conference Name: IEEE Transactions on Knowledge and Data Engineering.

- [60] Liu Yang, Di Chai, Junxue Zhang, Yilun Jin, Leye Wang, Hao Liu, Han Tian, Qian Xu, and Kai Chen. A Survey on Vertical Federated Learning: From a Layered Perspective, April 2023. URL <http://arxiv.org/abs/2304.01829>. arXiv:2304.01829 [cs].
- [61] Mang Ye, Wei Shen, Bo Du, Eduard Snezhko, Vassili Kovalev, and Pong C. Yuen. Vertical Federated Learning for Effectiveness, Security, Applicability: A Survey, June 2024. URL <http://arxiv.org/abs/2405.17495>. arXiv:2405.17495 [cs].
- [62] Wei Guo, Fuzhen Zhuang, Xiao Zhang, Yiqi Tong, and Jin Dong. A comprehensive survey of federated transfer learning: challenges, methods and applications. *Frontiers of Computer Science*, 18(6):186356, July 2024. ISSN 2095-2236. doi: 10.1007/s11704-024-40065-x. URL <https://doi.org/10.1007/s11704-024-40065-x>.
- [63] Quan Qian, Bin Zhang, Chuan Li, Yongfang Mao, and Yi Qin. Federated transfer learning for machinery fault diagnosis: A comprehensive review of technique and application. *Mechanical Systems and Signal Processing*, 223:111837, January 2025. ISSN 0888-3270. doi: 10.1016/j.ymssp.2024.111837. URL <https://www.sciencedirect.com/science/article/pii/S0888327024007350>.
- [64] Seunghyeon Shin, Mallika Boyapati, Kun Suo, Kyungtae Kang, and Junggab Son. An empirical analysis of image augmentation against model inversion attack in federated learning. *Cluster Computing*, 26(1):349–366, February 2023. ISSN 1573-7543. doi: 10.1007/s10586-022-03596-1. URL <https://doi.org/10.1007/s10586-022-03596-1>.
- [65] Wael Issa, Nour Moustafa, Benjamin Turnbull, and Kim-Kwang Raymond Choo. RVE-PFL: Robust Variational Encoder-Based Personalized Federated Learning Against Model Inversion Attacks. *IEEE Transactions on Information Forensics and Security*, 19:3772–3787, 2024. ISSN 1556-6021. doi: 10.1109/TIFS.2024.3368879. URL <https://ieeexplore.ieee.org/document/10443672/?arnumber=10443672>. Conference Name: IEEE Transactions on Information Forensics and Security.
- [66] Shanghao Shi, Ning Wang, Yang Xiao, Chaoyu Zhang, Yi Shi, Y. Thomas Hou, and Wenjing Lou. Scale-MIA: A Scalable Model Inversion Attack against Secure Federated Learning via

Latent Space Reconstruction, November 2024. URL <http://arxiv.org/abs/2311.05808>. arXiv:2311.05808 [cs].

- [67] Jingtao Li, Adnan Siraj Rakin, Xing Chen, Zhezhi He, Deliang Fan, and Chaitali Chakrabarti. ResSFL: A Resistance Transfer Framework for Defending Model Inversion Attack in Split Federated Learning. pages 10194–10202, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Li_ResSFL_A_Resistance_Transfer_Framework_for_Defending_Model_Inversion_Attack_CVPR_2022_paper.html.
- [68] Jonas Geiping, Hartmut Bauermeister, Hannah Dröge, and Michael Moeller. Inverting Gradients - How easy is it to break privacy in federated learning? In *Advances in Neural Information Processing Systems*, volume 33, pages 16937–16947. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/c4ede56bbd98819ae6112b20ac6bf145-Abstract.html>.
- [69] Ali Hatamizadeh, Hongxu Yin, Pavlo Molchanov, Andriy Myronenko, Wenqi Li, Prerna Dogra, Andrew Feng, Mona G Flores, Jan Kautz, Daguang Xu, and Holger R. Roth. Do Gradient Inversion Attacks Make Federated Learning Unsafe? *IEEE Transactions on Medical Imaging*, 42(7):2044–2056, July 2023. ISSN 1558-254X. doi: 10.1109/TMI.2023.3239391. URL <https://ieeexplore.ieee.org/document/10025466/?arnumber=10025466>. Conference Name: IEEE Transactions on Medical Imaging.
- [70] Hongsheng Hu, Zoran Salcic, Lichao Sun, Gillian Dobbie, and Xuyun Zhang. Source Inference Attacks in Federated Learning. In *2021 IEEE International Conference on Data Mining (ICDM)*, pages 1102–1107, December 2021. doi: 10.1109/ICDM51629.2021.00129. URL <https://ieeexplore.ieee.org/document/9679121/?arnumber=9679121>. ISSN: 2374-8486.
- [71] Ligeng Zhu, Zhijian Liu, and Song Han. Deep Leakage from Gradients. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/60a6c4002cc7b29142def8871531281a-Abstract.html>.
- [72] Zhibo Wang, Mengkai Song, Zhifei Zhang, Yang Song, Qian Wang, and Hairong Qi. Beyond Inferring Class Representatives: User-Level Privacy Leakage From Federated Learning. In *IEEE INFOCOM 2019 - IEEE Conference on Computer Communications*, pages 2512–2520,

April 2019. doi: 10.1109/INFOCOM.2019.8737416. URL <https://ieeexplore.ieee.org/document/8737416/?arnumber=8737416>. ISSN: 2641-9874.

- [73] Oualid Zari, Chuan Xu, and Giovanni Neglia. Efficient passive membership inference attack in federated learning, October 2021. URL <http://arxiv.org/abs/2111.00430>. arXiv:2111.00430 [cs].
- [74] Li Bai, Haibo Hu, Qingqing Ye, Haoyang Li, Leixia Wang, and Jianliang Xu. Membership Inference Attacks and Defenses in Federated Learning: A Survey. *ACM Comput. Surv.*, 57(4):89:1–89:35, December 2024. ISSN 0360-0300. doi: 10.1145/3704633. URL <https://dl.acm.org/doi/10.1145/3704633>.
- [75] Anshuman Suri, Pallika Kanani, Virendra J. Marathe, and Daniel W. Peterson. Subject Membership Inference Attacks in Federated Learning, June 2023. URL <http://arxiv.org/abs/2206.03317>. arXiv:2206.03317 [cs].
- [76] Stacey Truex, Ling Liu, Mehmet Emre Gursay, Lei Yu, and Wenqi Wei. Towards Demystifying Membership Inference Attacks, February 2019. URL <http://arxiv.org/abs/1807.09173>. arXiv:1807.09173 [cs].
- [77] Xinlong He, Yang Xu, Sicong Zhang, Weida Xu, and Jiale Yan. Enhance membership inference attacks in federated learning. *Computers & Security*, 136:103535, January 2024. ISSN 0167-4048. doi: 10.1016/j.cose.2023.103535. URL <https://www.sciencedirect.com/science/article/pii/S0167404823004455>.
- [78] Xinjian Luo and Xianglong Zhang. Exploiting Defenses against GAN-Based Feature Inference Attacks in Federated Learning, August 2024. URL <http://arxiv.org/abs/2004.12571>. arXiv:2004.12571 [cs].
- [79] Rohit Mapakshi. An Empirical Analysis of Adversarial Attacks in Federated Learning. *Master's Projects*, January 2024. URL https://scholarworks.sjsu.edu/etd_projects/1422.
- [80] Briland Hitaj, Giuseppe Ateniese, and Fernando Perez-Cruz. Deep Models Under the GAN: Information Leakage from Collaborative Deep Learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, CCS '17, pages 603–618,

- New York, NY, USA, October 2017. Association for Computing Machinery. ISBN 978-1-4503-4946-8. doi: 10.1145/3133956.3134012. URL <https://doi.org/10.1145/3133956.3134012>.
- [81] Shuo Xu, Hui Xia, Lijuan Xu, Rui Zhang, and Chunqiang Hu. MIGAN: A Privacy Leakage Evaluation Scheme for CIoT-Based Federated Learning Users. *IEEE Transactions on Consumer Electronics*, 70(1):3098–3110, February 2024. ISSN 1558-4127. doi: 10.1109/TCE.2024.3367494. URL <https://ieeexplore.ieee.org/document/10440139/?arnumber=10440139>. Conference Name: IEEE Transactions on Consumer Electronics.
- [82] Jingwen Zhang, Jiale Zhang, Junjun Chen, and Shui Yu. GAN Enhanced Membership Inference: A Passive Local Attack in Federated Learning. In *ICC 2020 - 2020 IEEE International Conference on Communications (ICC)*, pages 1–6, June 2020. doi: 10.1109/ICC40277.2020.9148790. URL <https://ieeexplore.ieee.org/document/9148790/?arnumber=9148790>. ISSN: 1938-1883.
- [83] Prajjwal Gupta, Krishna Yadav, Brij B. Gupta, Mamoun Alazab, and Thippa Reddy Gadekallu. A Novel Data Poisoning Attack in Federated Learning based on Inverted Loss Function. *Computers & Security*, 130:103270, July 2023. ISSN 0167-4048. doi: 10.1016/j.cose.2023.103270. URL <https://www.sciencedirect.com/science/article/pii/S0167404823001803>.
- [84] Gan Sun, Yang Cong, Jiahua Dong, Qiang Wang, Lingjuan Lyu, and Ji Liu. Data Poisoning Attacks on Federated Machine Learning. *IEEE Internet of Things Journal*, 9(13):11365–11375, July 2022. ISSN 2327-4662. doi: 10.1109/JIOT.2021.3128646. URL <https://ieeexplore.ieee.org/document/9618642/?arnumber=9618642>. Conference Name: IEEE Internet of Things Journal.
- [85] Florian Nuding and Rudolf Mayer. Data Poisoning in Sequential and Parallel Federated Learning. In *Proceedings of the 2022 ACM on International Workshop on Security and Privacy Analytics, IWSPA '22*, pages 24–34, New York, NY, USA, April 2022. Association for Computing Machinery. ISBN 978-1-4503-9230-3. doi: 10.1145/3510548.3519372. URL <https://dl.acm.org/doi/10.1145/3510548.3519372>.
- [86] Konstantinos Psychogyios, Terpsichori-Helen Velivassaki, Stavroula Bourou, Artemis Voulkidis, Dimitrios Skias, and Theodore Zahariadis. GAN-Driven Data Poisoning At-

- tacks and Their Mitigation in Federated Learning Systems. *Electronics*, 12(8):1805, January 2023. ISSN 2079-9292. doi: 10.3390/electronics12081805. URL <https://www.mdpi.com/2079-9292/12/8/1805>. Number: 8 Publisher: Multidisciplinary Digital Publishing Institute.
- [87] Yao Chen, Yijie Gui, Hong Lin, Wensheng Gan, and Yongdong Wu. Federated Learning Attacks and Defenses: A Survey. In *2022 IEEE International Conference on Big Data (Big Data)*, pages 4256–4265, December 2022. doi: 10.1109/BigData55660.2022.10020431. URL <https://ieeexplore.ieee.org/document/10020431/?arnumber=10020431>.
- [88] Kummari Naveen Kumar, Chalavadi Krishna Mohan, and Linga Reddy Cenkeramaddi. The Impact of Adversarial Attacks on Federated Learning: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):2672–2691, May 2024. ISSN 1939-3539. doi: 10.1109/TPAMI.2023.3322785. URL <https://ieeexplore.ieee.org/document/10274102/?arnumber=10274102>. Conference Name: IEEE Transactions on Pattern Analysis and Machine Intelligence.
- [89] Hira Shahzadi Sikandar, Huda Waheed, Sibgha Tahir, Saif U. R. Malik, and Waqas Rafique. A Detailed Survey on Federated Learning Attacks and Defenses. *Electronics*, 12(2):260, January 2023. ISSN 2079-9292. doi: 10.3390/electronics12020260. URL <https://www.mdpi.com/2079-9292/12/2/260>. Number: 2 Publisher: Multidisciplinary Digital Publishing Institute.
- [90] Geming Xia, Jian Chen, Chaodong Yu, and Jun Ma. Poisoning Attacks in Federated Learning: A Survey. *IEEE Access*, 11:10708–10722, 2023. ISSN 2169-3536. doi: 10.1109/ACCESS.2023.3238823. URL <https://ieeexplore.ieee.org/document/10024252/?arnumber=10024252>. Conference Name: IEEE Access.
- [91] Xueluan Gong, Yanjiao Chen, Qian Wang, and Weihang Kong. Backdoor Attacks and Defenses in Federated Learning: State-of-the-Art, Taxonomy, and Future Directions. *IEEE Wireless Communications*, 30(2):114–121, April 2023. ISSN 1558-0687. doi: 10.1109/MWC.017.2100714. URL <https://ieeexplore.ieee.org/document/9806416/?arnumber=9806416>. Conference Name: IEEE Wireless Communications.
- [92] Xianghua Xie, Chen Hu, Hanchi Ren, and Jingjing Deng. A survey on vulnerability of federated learning: A learning algorithm perspective. *Neurocomputing*, 573:127225, March 2024.

ISSN 0925-2312. doi: 10.1016/j.neucom.2023.127225. URL <https://www.sciencedirect.com/science/article/pii/S0925231223013486>.

- [93] Shenghui Li, Edith C.-H. Ngai, and Thiemo Voigt. An Experimental Study of Byzantine-Robust Aggregation Schemes in Federated Learning. *IEEE Transactions on Big Data*, 10(6):975–988, December 2024. ISSN 2332-7790. doi: 10.1109/TBDATA.2023.3237397. URL <https://ieeexplore.ieee.org/document/10018261/?arnumber=10018261>. Conference Name: IEEE Transactions on Big Data.
- [94] Marco Arazzi, Mauro Conti, Antonino Nocera, and Stjepan Picek. Turning Privacy-preserving Mechanisms against Federated Learning. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security, CCS '23*, pages 1482–1495, New York, NY, USA, November 2023. Association for Computing Machinery. ISBN 979-8-4007-0050-7. doi: 10.1145/3576915.3623114. URL <https://doi.org/10.1145/3576915.3623114>.
- [95] Aneel Sharma and Ningrui Marchang. Probabilistic Sign Flipping Attack in Federated Learning. In *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pages 1–6, June 2024. doi: 10.1109/ICCCNT61001.2024.10725463. URL <https://ieeexplore.ieee.org/document/10725463/?arnumber=10725463>. ISSN: 2473-7674.
- [96] Cong Xie, Oluwasanmi Koyejo, and Indranil Gupta. Fall of Empires: Breaking Byzantine-tolerant SGD by Inner Product Manipulation. In *Proceedings of The 35th Uncertainty in Artificial Intelligence Conference*, pages 261–270. PMLR, August 2020. URL <https://proceedings.mlr.press/v115/xie20a.html>. ISSN: 2640-3498.
- [97] Yanbin Li, Xuemei Wang, Runkang Sun, Xiaojun Xie, Shijia Ying, and Shougang Ren. Trustiness-based hierarchical decentralized federated learning. *Knowledge-Based Systems*, 276:110763, September 2023. ISSN 0950-7051. doi: 10.1016/j.knosys.2023.110763. URL <https://www.sciencedirect.com/science/article/pii/S0950705123005130>.
- [98] Jie Yang, Thar Baker, Sukhpal Singh Gill, Xiaochuan Yang, Weifeng Han, and Yuanzhang Li. A federated learning attack method based on edge collaboration via cloud. *Software: Practice and Experience*, 54(7):1257–1274, 2024. ISSN 1097-024X. doi: 10.1002/

- spe.3180. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/spe.3180>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/spe.3180>.
- [99] Gilad Baruch, Moran Baruch, and Yoav Goldberg. A Little Is Enough: Circumventing Defenses For Distributed Learning. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/ec1c59141046cd1866bbbcdfb6ae31d4-Abstract.html.
- [100] Tongsai Jin, Zhihui Fu, Dan Meng, Jun Wang, Yue Qi, and Guitao Cao. FedPerturb: Covert Poisoning Attack on Federated Learning via Partial Perturbation. In *ECAI 2023*, pages 1172–1179. IOS Press, 2023. doi: 10.3233/FAIA230393. URL <https://ebooks.iospress.nl/doi/10.3233/FAIA230393>.
- [101] Xiaoyu Cao, Minghong Fang, Jia Liu, and Neil Zhenqiang Gong. FLTrust: Byzantine-robust Federated Learning via Trust Bootstrapping, April 2022. URL <http://arxiv.org/abs/2012.13995>. arXiv:2012.13995 [cs].
- [102] Moming Duan, Duo Liu, Xinyuan Ji, Renping Liu, Liang Liang, Xianzhang Chen, and Yujian Tan. FedGroup: Efficient Federated Learning via Decomposed Similarity-Based Clustering. In *2021 IEEE Intl Conf on Parallel & Distributed Processing with Applications, Big Data & Cloud Computing, Sustainable Computing & Communications, Social Computing & Networking (ISPA/BDCLOUD/SocialCom/SustainCom)*, pages 228–237, September 2021. doi: 10.1109/ISPA-BDCLOUD-SocialCom-SustainCom52081.2021.00042. URL <https://ieeexplore.ieee.org/abstract/document/9644782>.
- [103] Najeeb Moharram Jebreel and Josep Domingo-Ferrer. FL-Defender: Combating targeted attacks in federated learning. *Knowledge-Based Systems*, 260:110178, January 2023. ISSN 0950-7051. doi: 10.1016/j.knosys.2022.110178. URL <https://www.sciencedirect.com/science/article/pii/S0950705122012746>.
- [104] Peva Blanchard, El Mahdi El Mhamdi, Rachid Guerraoui, and Julien Stainer. Machine Learning with Adversaries: Byzantine Tolerant Gradient Descent. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/f4b9ec30ad9f68f89b29639786cb62ef-Abstract.html>.

- [105] Dong Yin, Yudong Chen, Ramchandran Kannan, and Peter Bartlett. Byzantine-Robust Distributed Learning: Towards Optimal Statistical Rates. In *Proceedings of the 35th International Conference on Machine Learning*, pages 5650–5659. PMLR, July 2018. URL <https://proceedings.mlr.press/v80/yin18a.html>. ISSN: 2640-3498.
- [106] Sai Praneeth Karimireddy, Lie He, and Martin Jaggi. Learning from History for Byzantine Robust Optimization. In *Proceedings of the 38th International Conference on Machine Learning*, pages 5311–5319. PMLR, July 2021. URL <https://proceedings.mlr.press/v139/karimireddy21a.html>. ISSN: 2640-3498.
- [107] Wen Huang, Ming Zhuo, Tianqing Zhu, Shijie Zhou, and Yongjian Liao. Differential privacy: Review of improving utility through cryptography-based technologies. *Concurrency and Computation: Practice and Experience*, 35(5):e7565, 2023. ISSN 1532-0634. doi: 10.1002/cpe.7565. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/cpe.7565>. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/cpe.7565>.
- [108] Cynthia Dwork, Nitin Kohli, and Deirdre Mulligan. Differential Privacy in Practice: Expose your Epsilons! *Journal of Privacy and Confidentiality*, 9(2), 2019. ISSN 2575-8527. doi: 10.29012/jpc.689. URL <https://escholarship.org/uc/item/8vj5q8j7>.
- [109] Kenneth Goss. Combining Shamir & Additive Secret Sharing to Improve Efficiency of SMC Primitives Against Malicious Adversaries, November 2023. URL <http://arxiv.org/abs/2311.08934>. arXiv:2311.08934 [cs].
- [110] Yiwei Zhang, Rouzbeh Behnia, Attila A. Yavuz, Reza Ebrahimi, and Elisa Bertino. Uncovering Attacks and Defenses in Secure Aggregation for Federated Deep Learning, October 2024. URL <http://arxiv.org/abs/2410.09676>. arXiv:2410.09676 [cs].
- [111] Saad Sajid Hashmi, Hoa Khanh Dam, Peter Smet, and Mohan Baruwal Chhetri. Towards Antifragility in Contested Environments: Using Adversarial Search to Learn, Predict, and Counter Open-Ended Threats. In *2022 IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS)*, pages 141–146, September 2022. doi: 10.1109/ACSOS55765.2022.00032. URL <https://ieeexplore.ieee.org/document/9935020/?arnumber=9935020>.

- [112] Adnan Qayyum, Aneeqa Ijaz, Muhammad Usama, Waleed Iqbal, Junaid Qadir, Yehia Elkhatib, and Ala Al-Fuqaha. Securing Machine Learning in the Cloud: A Systematic Review of Cloud Machine Learning Security. *Frontiers in Big Data*, 3, November 2020. ISSN 2624-909X. doi: 10.3389/fdata.2020.587139. URL <https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2020.587139/full>. Publisher: Frontiers.
- [113] Eugene Bagdasaryan, Andreas Veit, Yiqing Hua, Deborah Estrin, and Vitaly Shmatikov. How To Backdoor Federated Learning. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, pages 2938–2948. PMLR, June 2020. URL <https://proceedings.mlr.press/v108/bagdasaryan20a.html>. ISSN: 2640-3498.
- [114] Christopher Frederickson, Michael Moore, Glenn Dawson, and Robi Polikar. Attack Strength vs. Detectability Dilemma in Adversarial Machine Learning. In *2018 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, July 2018. doi: 10.1109/IJCNN.2018.8489495. URL <https://ieeexplore.ieee.org/abstract/document/8489495>. ISSN: 2161-4407.
- [115] Diabetes, Hypertension and Stroke Prediction, 2023. URL <https://www.kaggle.com/datasets/prosperchuks/health-dataset>.
- [116] Virat Shejwalkar and Amir Houmansadr. Manipulating the Byzantine: Optimizing Model Poisoning Attacks and Defenses for Federated Learning. In *Proceedings 2021 Network and Distributed System Security Symposium*, Virtual, 2021. Internet Society. ISBN 978-1-891562-66-2. doi: 10.14722/ndss.2021.24498. URL https://www.ndss-symposium.org/wp-content/uploads/ndss2021_6C-3_24498_paper.pdf.
- [117] Ning Wang, Yang Xiao, Yimin Chen, Yang Hu, Wenjing Lou, and Y. Thomas Hou. FLARE: Defending Federated Learning against Model Poisoning Attacks via Latent Space Representations. In *Proceedings of the 2022 ACM on Asia Conference on Computer and Communications Security*, ASIA CCS '22, pages 946–958, New York, NY, USA, May 2022. Association for Computing Machinery. ISBN 978-1-4503-9140-5. doi: 10.1145/3488932.3517395. URL <https://dl.acm.org/doi/10.1145/3488932.3517395>.
- [118] Hend K. Gedawy, Khaled A. Harras, and Aiman Erbad. Leveraging Semi-Connected Devices To Enhance Federated Learning. In *2022 5th International Conference on Communications*,

- Signal Processing, and their Applications (ICCSPA)*, pages 1–6, December 2022. doi: 10.1109/ICCSPA55860.2022.10019249. URL <https://ieeexplore.ieee.org/document/10019249/?arnumber=10019249>. ISSN: 2767-7702.
- [119] Liangkun Yu, Xiang Sun, Rana Albelaihi, and Chen Yi. Latency-Aware Semi-Synchronous Client Selection and Model Aggregation for Wireless Federated Learning. *Future Internet*, 15(11):352, November 2023. ISSN 1999-5903. doi: 10.3390/fi15110352. URL <https://www.mdpi.com/1999-5903/15/11/352>. Number: 11 Publisher: Multidisciplinary Digital Publishing Institute.
- [120] Maoxuan Yan, Qingcai Luo, Bo Zhang, and Shanbao Sun. Solving Client Dropout in Federated Learning via Client Similarity Discovery and Gradient Supplementation Mechanism. In Zahir Tari, Keqiu Li, and Hongyi Wu, editors, *Algorithms and Architectures for Parallel Processing*, pages 446–457, Singapore, 2024. Springer Nature. ISBN 978-981-97-0808-6. doi: 10.1007/978-981-97-0808-6_26.
- [121] Xue Ying. An Overview of Overfitting and its Solutions. *Journal of Physics: Conference Series*, 1168(2):022022, February 2019. ISSN 1742-6596. doi: 10.1088/1742-6596/1168/2/022022. URL <https://dx.doi.org/10.1088/1742-6596/1168/2/022022>. Publisher: IOP Publishing.
- [122] Di Wu, Qi Guo, Yong Qi, Saiyu Qi, and Qian Li. AMA: Adaptive Model Poisoning Attacks towards Federated Learning. *IEEE Transactions on Dependable and Secure Computing*, pages 1–14, 2025. ISSN 1941-0018. doi: 10.1109/TDSC.2025.3594175. URL <https://ieeexplore.ieee.org/document/11108303/>.